GAZİ UNIVERSITY

FACULTY OF ENGINEERING
DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 3: DATABASE DESIGN AND CONCEPTUAL DESIGN

IMPORTANT ANNOUNCEMENT

- The course website is online now. Please check it out regularly.

- https://gazi-end-muh.github.io

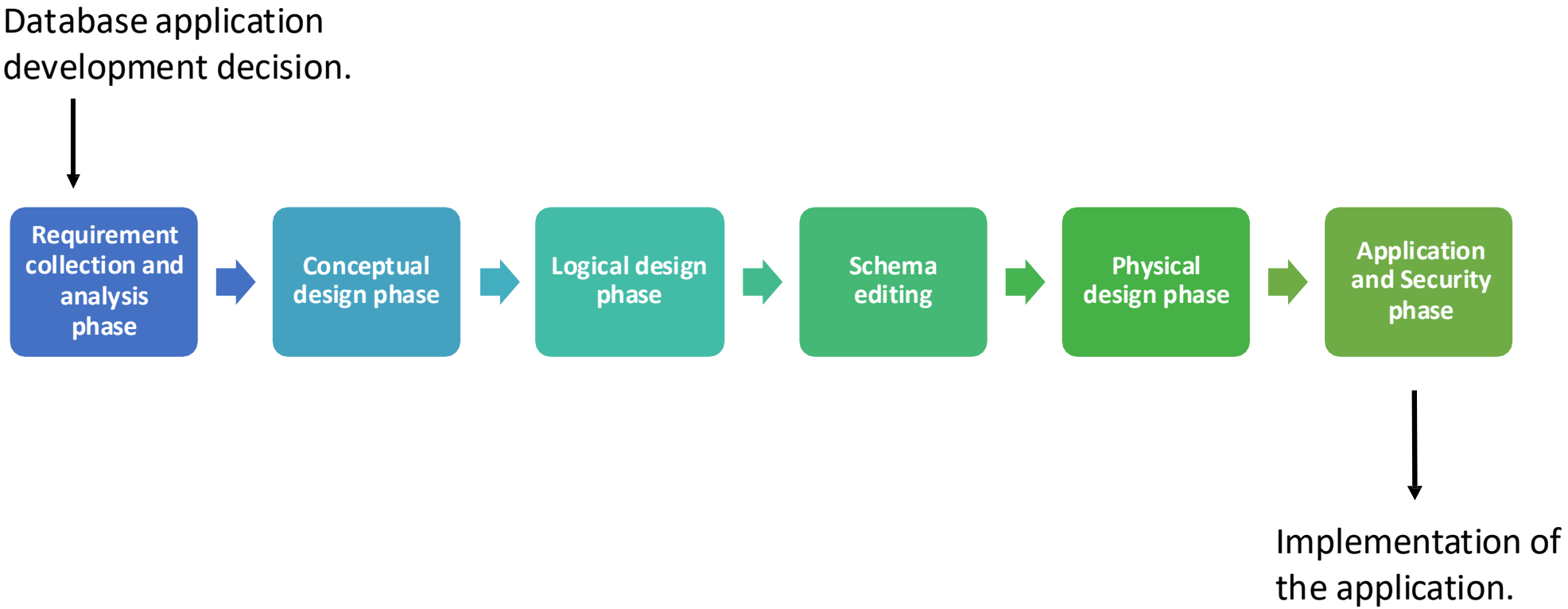
DATABASE DESIGN

WHAT IS NEEDED FOR DATABASE DESIGN?

- To develop database applications in a successful manner:

  - ☑ ***Collecting required resources***
  - ☑ ***Structured approach to project management.***
  - ☑ ***User training and user involvement.***
  - ☑ ***Team references.***
  - ☑ ***Rigorous technical specification.***
  - ☑ ***Similar software advantages and disadvantages.***
  - ☑ ***Product delivery and decision-making method.***
  - ☑ ***Senior management support.***

The Process of Database Design



1. Determination of Requirements

•**Importance of Definition:** A correct solution cannot be generated from an incorrectly defined problem; learning by trial-and-error is the most expensive method.

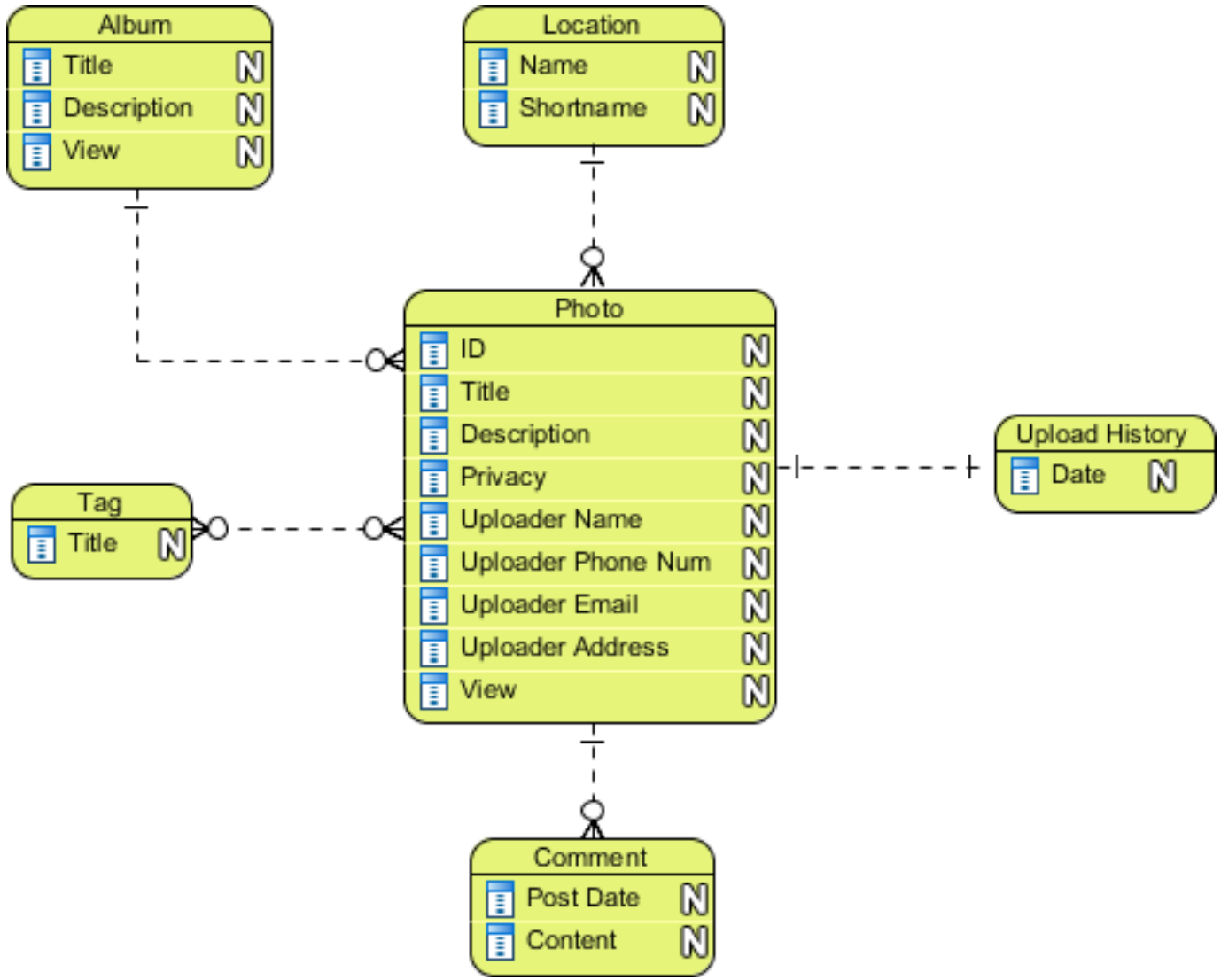
•**Team Formation:** A project team consisting of technical experts, end-users, and managers must be established at this stage.

•**Scope Consensus:** The team must agree on key elements: system users, data inputs, operational processes, outputs, and required reports.

•**Planning & Success:** A project schedule must be created, and specific success factors for all activities must be clearly defined.

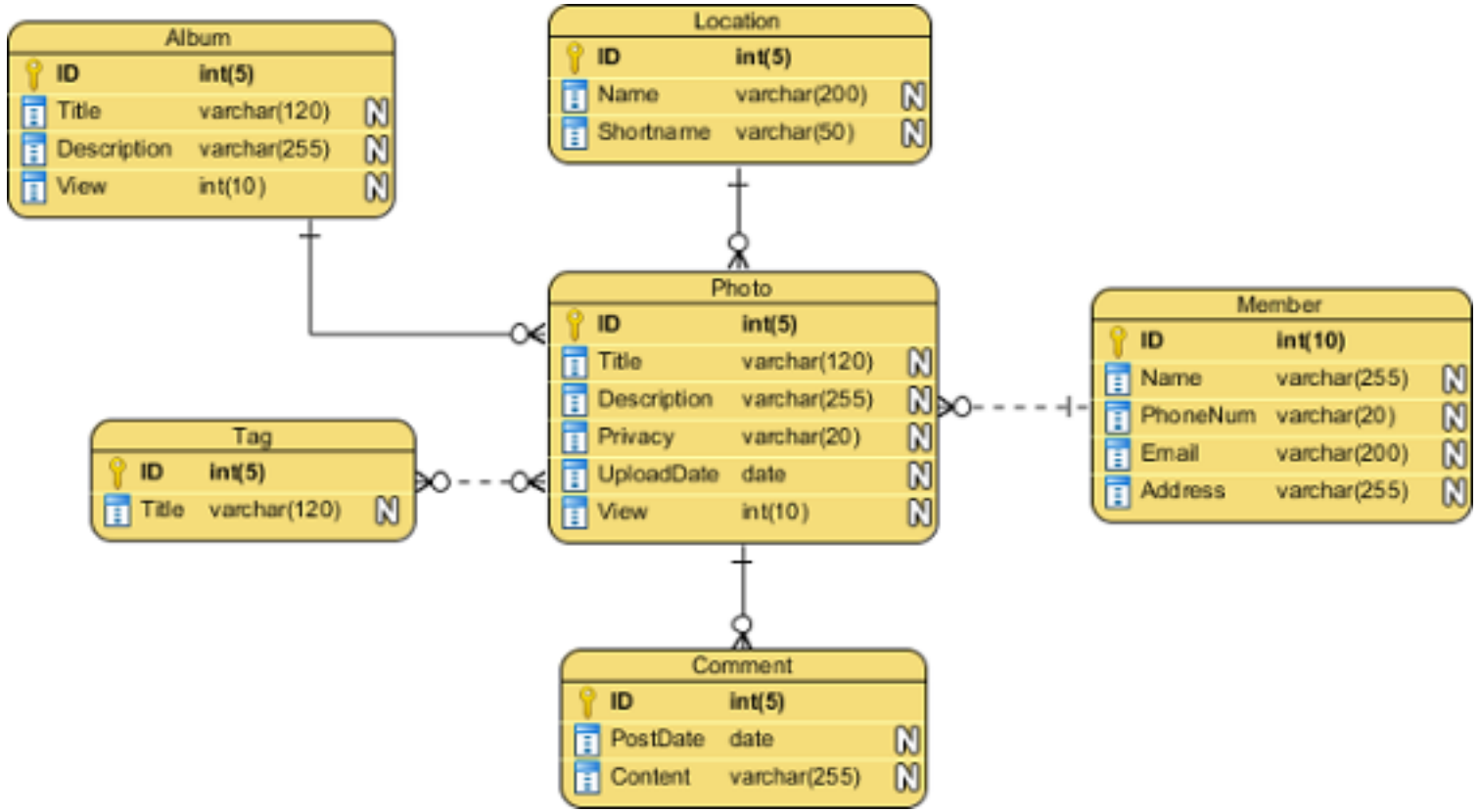
Conceptual Design

- It is the first stage the transformation of requirements into a working DB.
- It should be a concise design free from details.
- Data that will be stored in DB, relations between them, restrictions etc. should be expressed conceptually.
- The conceptual design should be capable of transforming into the data model to be used.



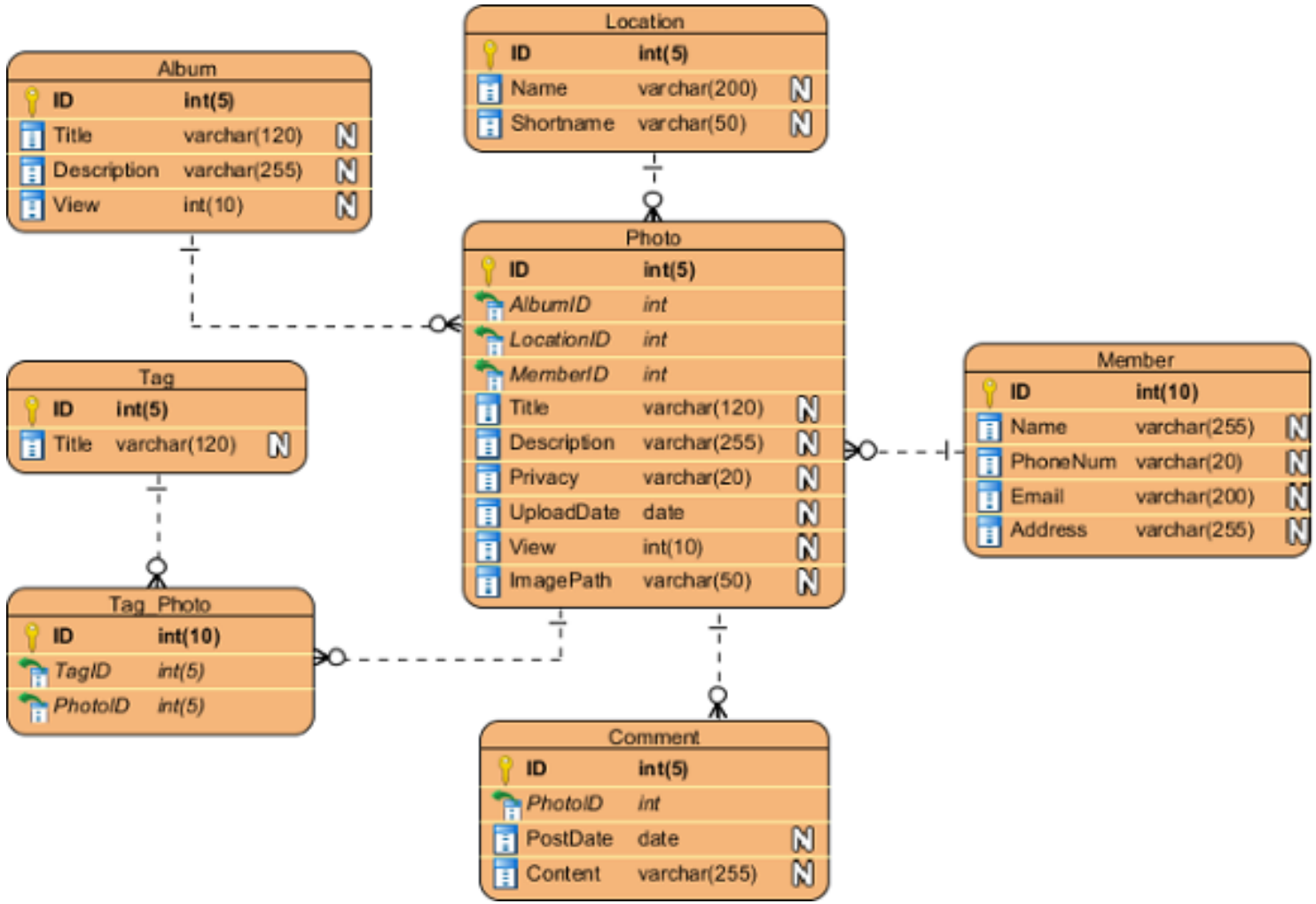
Logical Design

- A conceptual design can be converted into a database schemas in a structure, that is suitable for the data model to be used.
- The logical design should define a database that can work flawlessly in the DBMS tool to be used.



Physical Design

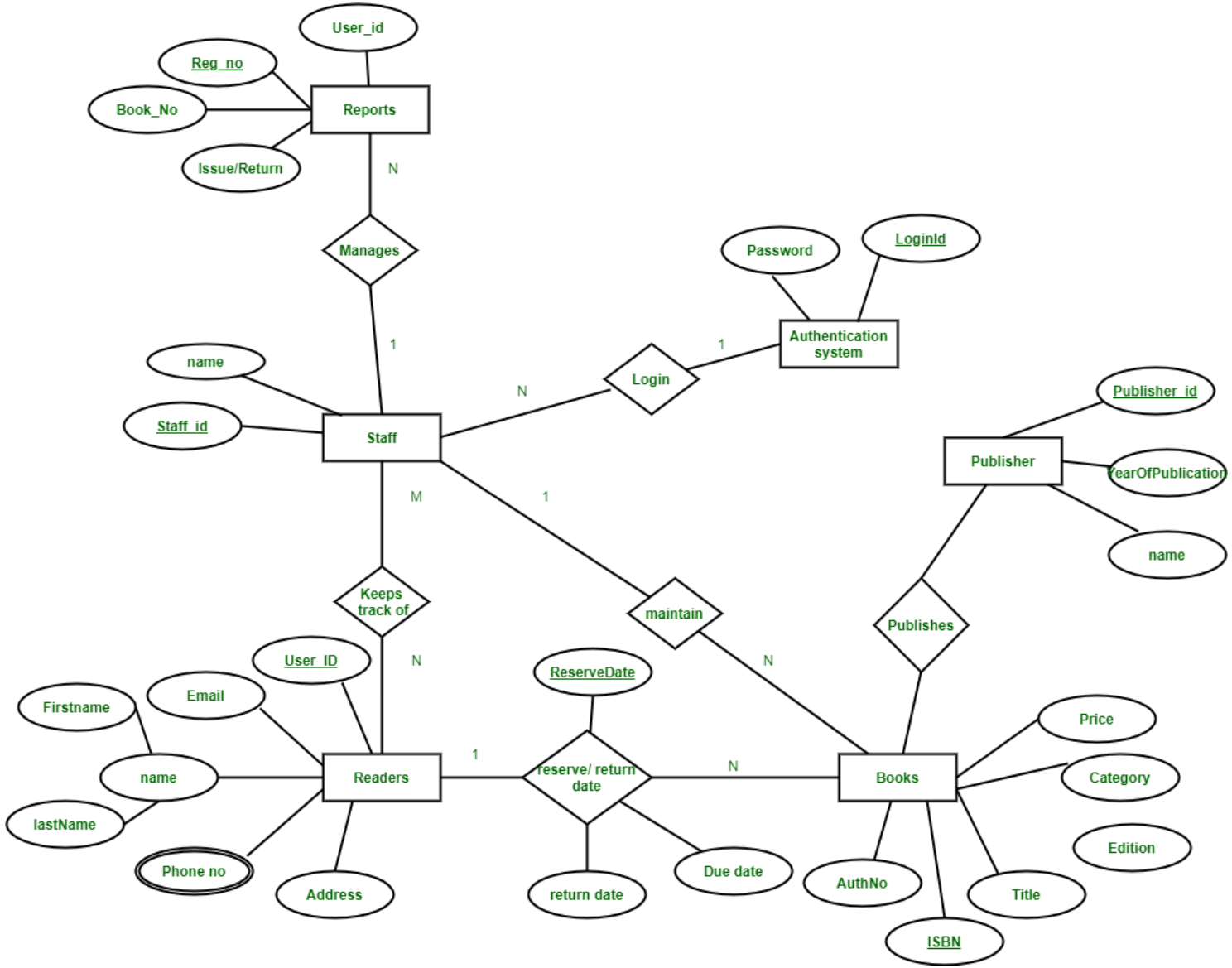
- It focuses on performance and space utilization.
- For this purpose, the features and components that the database should physically carry are determined and they are physically created.
- Such as; attribute design, physical register design, index design, buffer-oriented decisions, transaction design, etc.



CONCEPTUAL DESIGN

Objective

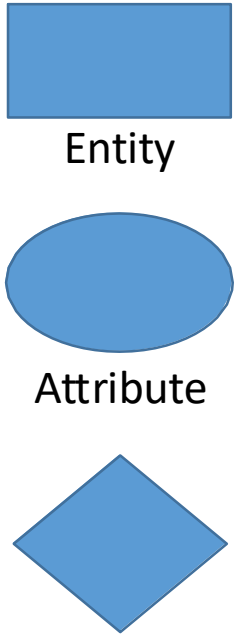
YOU ARE EXPECTED TO UNDERSTAND AND INTERPRET THE DIAGRAM SHOWN HERE AFTER THIS CLASS.



Introduction

For the Conceptual Design The ER method will be used.

- Most preferred tool
- Graphical
- Easy to learn
- Easy to use
- Effective



Microsoft SQL Server
ORACLE
PostgreSQL
MySQL

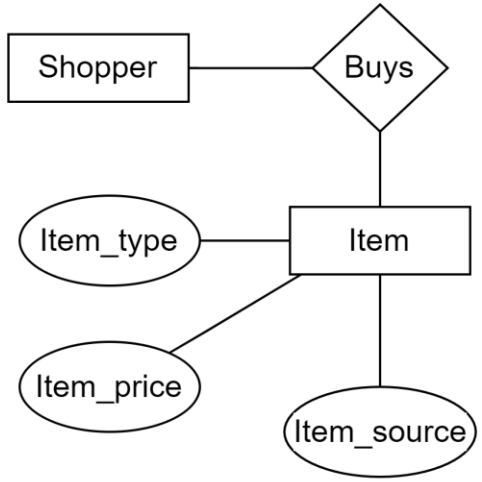
Introduction

The Output of System Analysis

Informal Information received from the user about what is expected from the system.

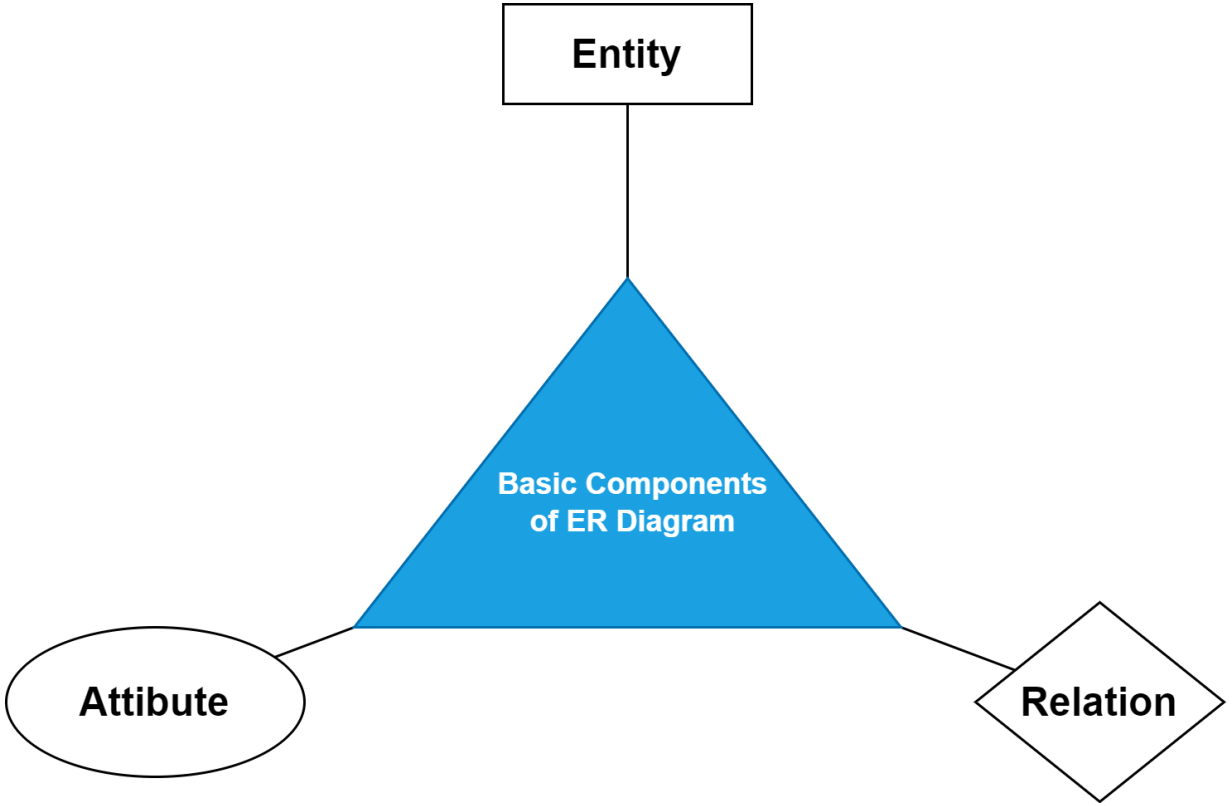
Conceptual Database Design with ER Model

- Ignores technical details.
- Easy to use
- Can be used to communicate with non technical person



- Graphical/Physical Representation
- Specific information to be used in the DB.
- Easily can be converted to a physical DB
- Easy to understand and modify
- Basic DB components and their interaction can be seen

Definitions

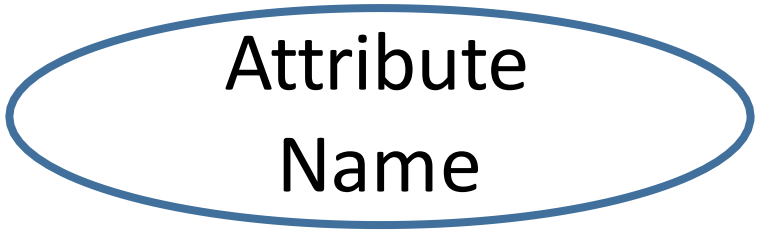


Entity

Entity Name

- Entity is one of the basic components of ER diagrams.
- Each object that can be distinguished from others in real life (personnel, student, product, etc.),
- Important perceptible elements (course, school, address, etc.)

Attribute

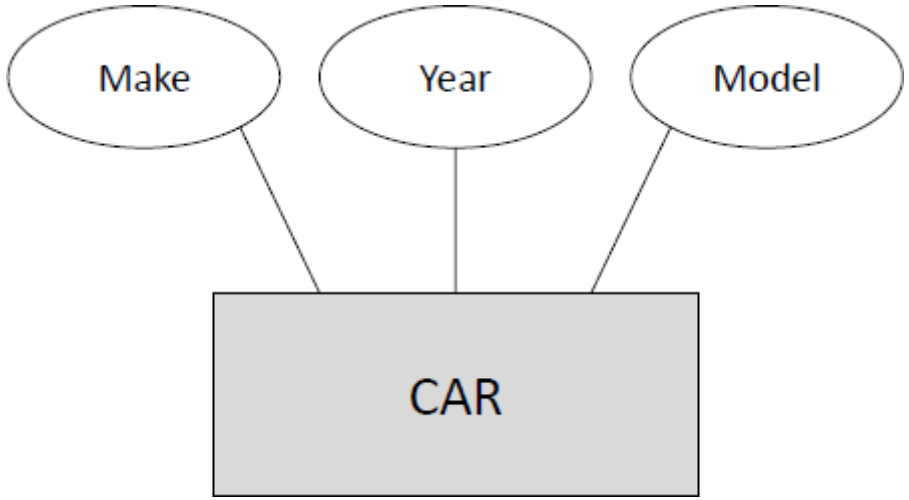


- Attributes are the characteristics or adjectives that define entities and distinguish them from other entities.
- Any number of attributes can be used for an entity.

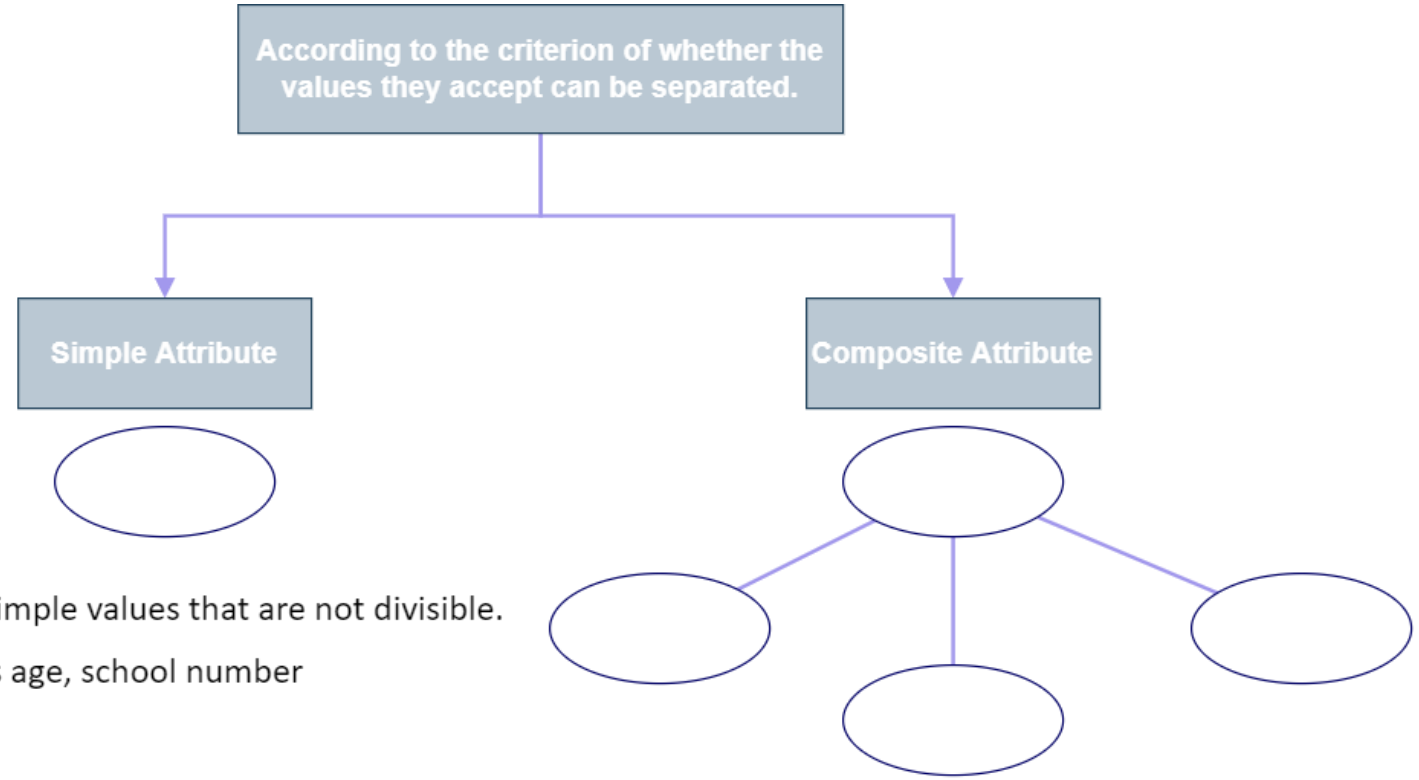
Entity	Possible Attributes
Student	Citizenship ID, Number, Name, Last Name, Birth Date…
Product	Code, Description, width, height, color etc.
Address	Neighborhood, Street, Door no, District, City, Zip code etc.
Class	Class number, floor, capacity, projection etc.

Attribute

- Set of entities that have the same attributes
- All cars have a year, make, and model.



Attribute Types

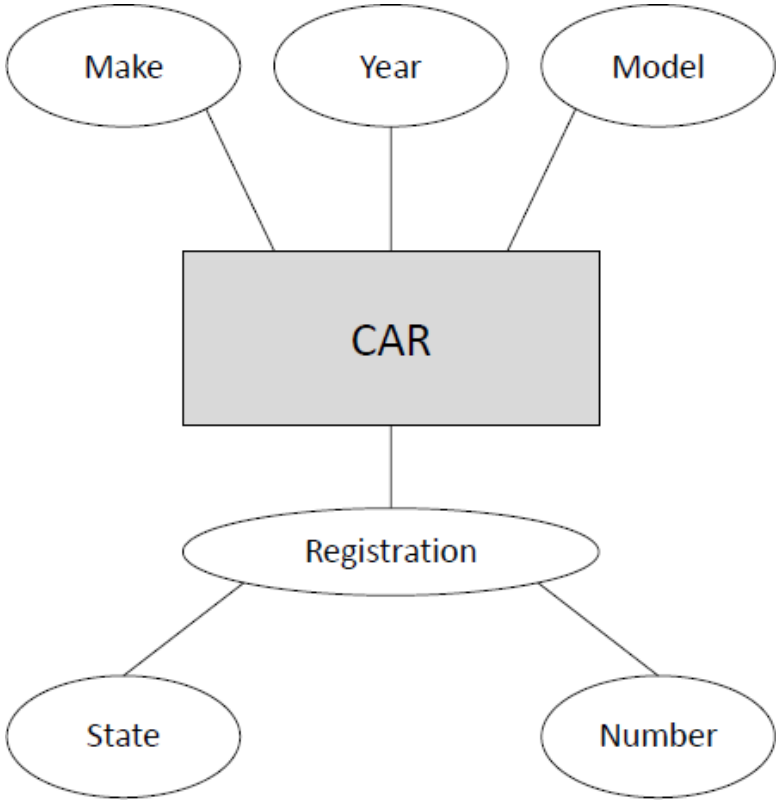


- Takes simple values that are not divisible.
- Such as age, school number

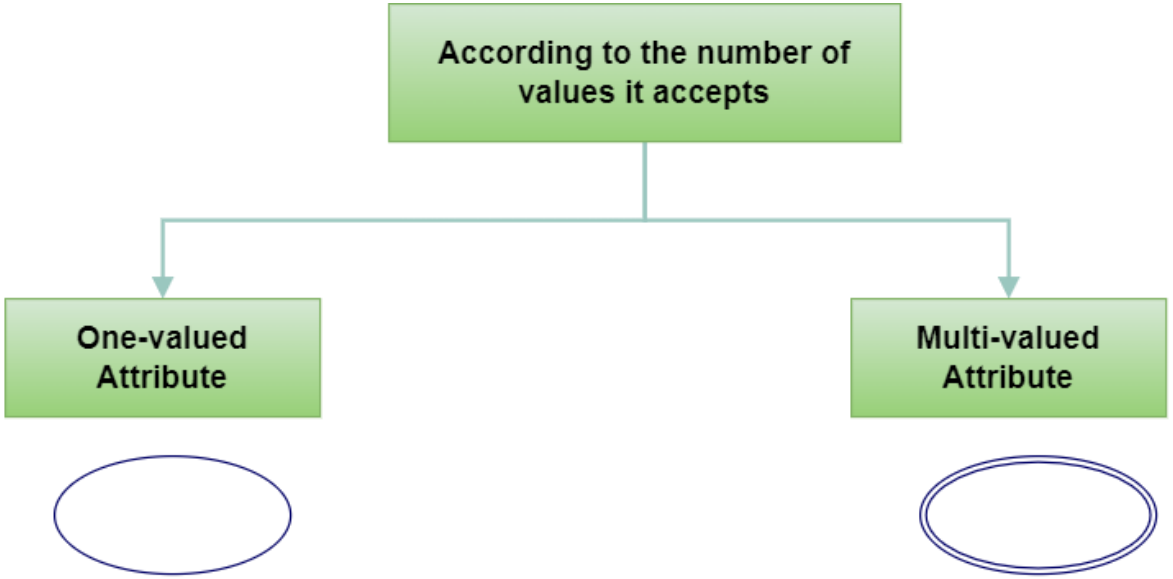
- Takes divisible values.
- There is a need for searching on the basis of subdivisions.
- Sub-attributes are repeated more than once.
- Such as address.

Composite Attributes

- Can be subdivided into smaller subparts
- All cars have a year, make, model, and **registration**.



Attribute Types

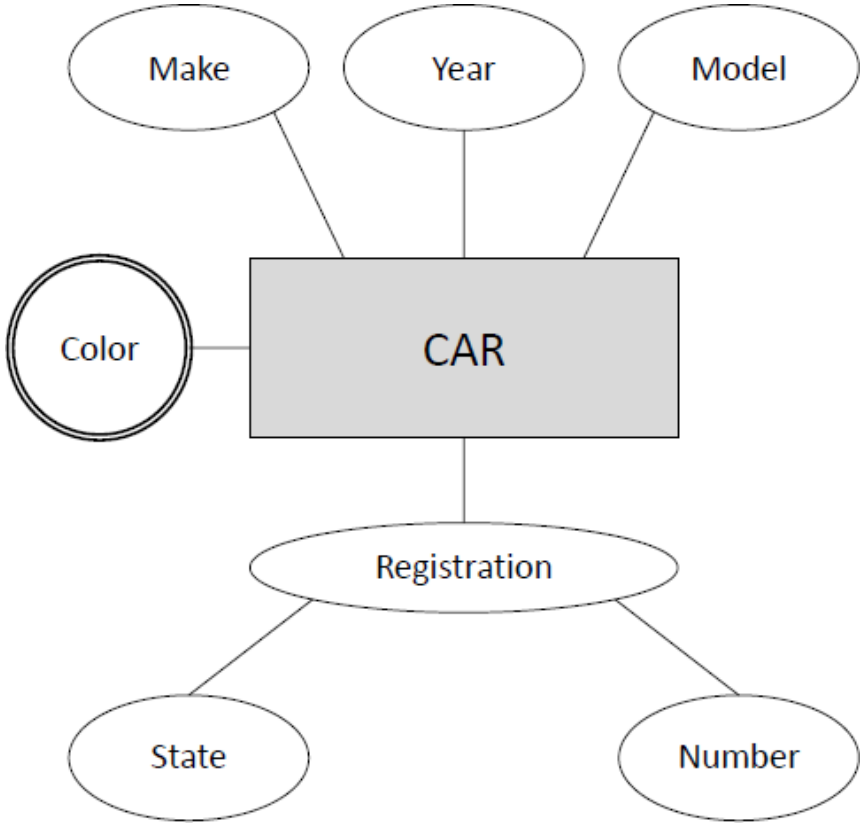


- Takes only one value.
- Such as age, last name, father name

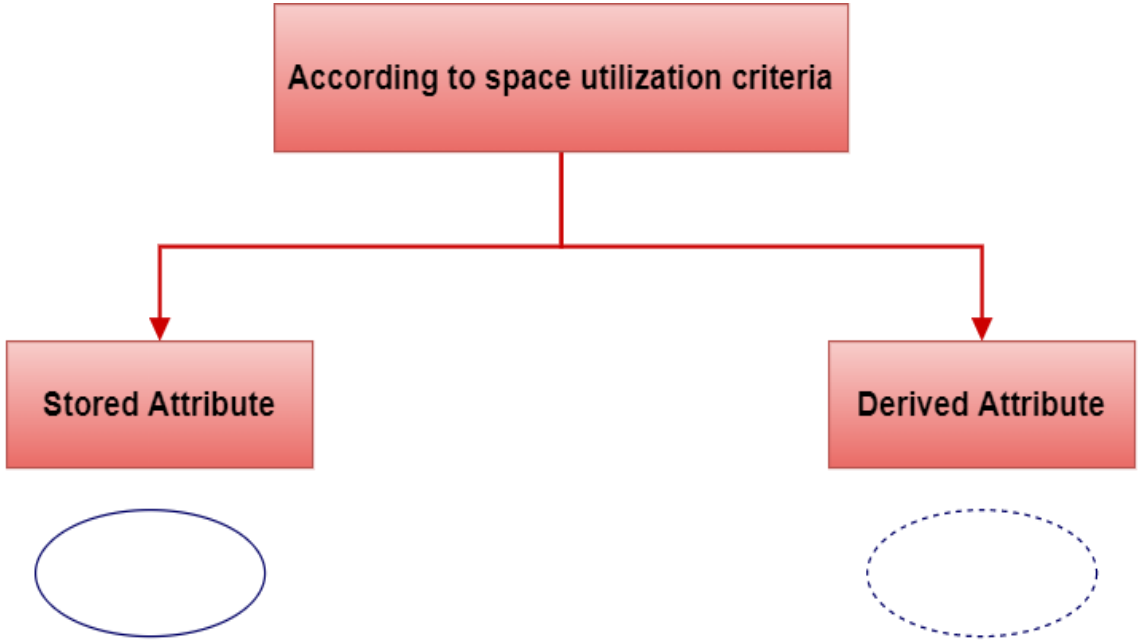
- a set of values from the same field is assigned.
- Such as certificate, product operation, usage area.

Multivalued Attributes

- Can take a [possibly specified] number of values.
- All cars have a year, make, model, registration, and some number of **colors**.



Attribute Types

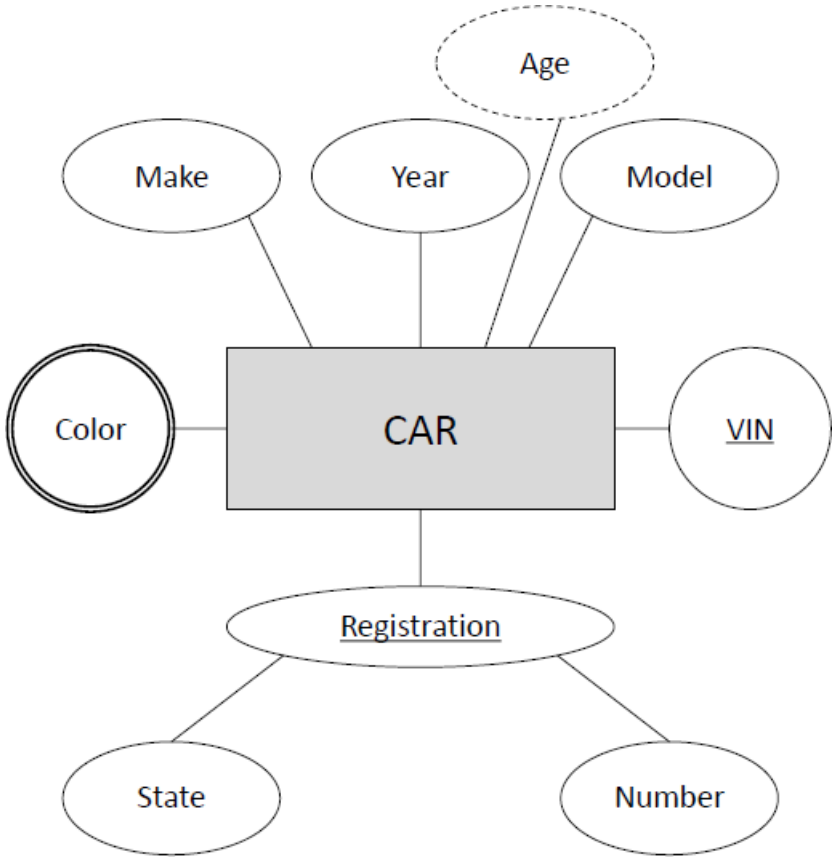


- Its value is stored in DB.
- Such age, last name, father name.

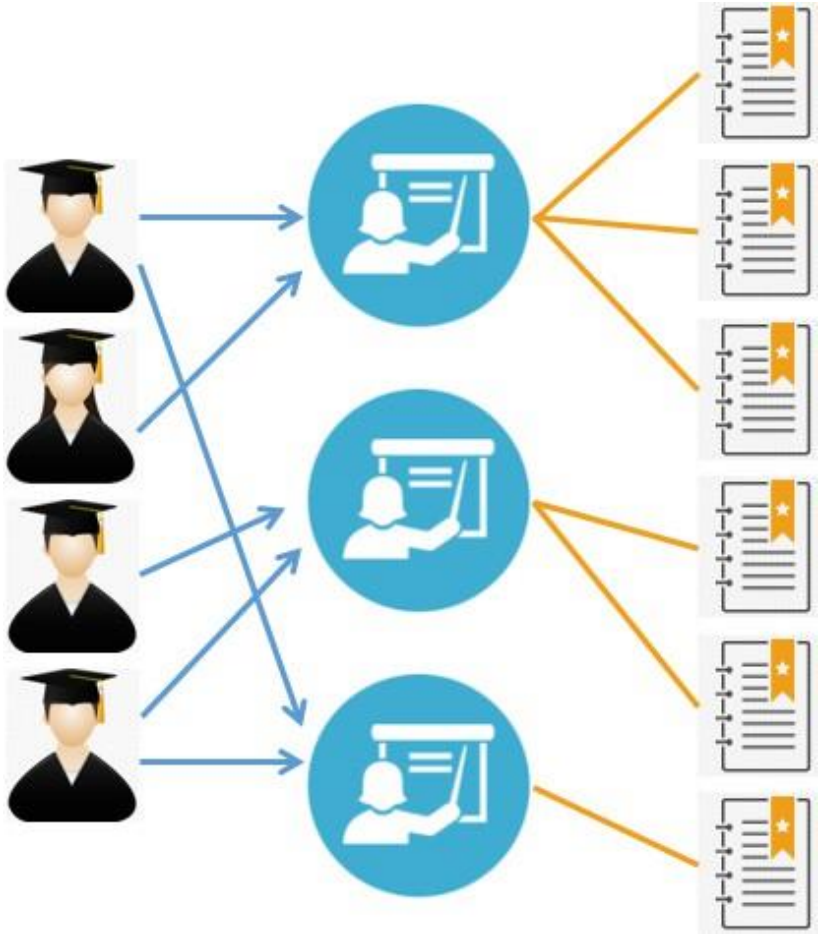
- Its value is calculated by one or more functions of the stored attributes.
- Such as total amount, number of students.

Derived Attributes

- The value can be computed.
- All cars have a year, **age**, make, model, registration (unique), vehicle identification number (vin; unique), some number of colors.



Relationships



- Relationship is the state of connecting entities to each other.
- Just like a student taking a course, a customer ordering.
- May have its own attributes. (Descriptive attributes.)
- It corresponds to a state, a predicate or an action between entities

Relationships

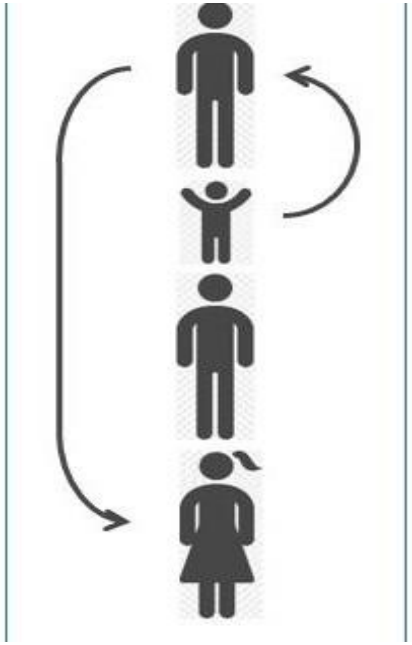
- Some entities and definitions of their relations.

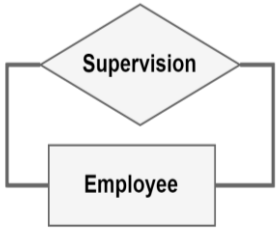
Entity-1	Entity-2	Relationship Description	Relationship Name
Student	Course	Student takes the course.	Take
Employee	Machine	Employee operates the machine.	Operate
Department	Fixture	Department has fixtures.	Has
Driver	Bus	Driver drives the bus.	Drive
Doctor	Patient	Doctor examines the patient.	Examine
Arbitrator	Project	Arbitrator evaluates the project.	Evaluate
Auditor	Company	Auditor inspects the company.	Inspect

Relationships

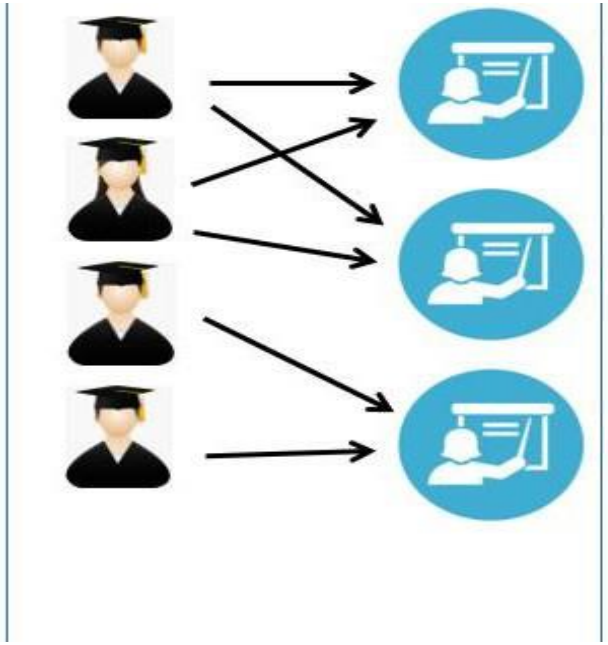
There exist a number of relationship types

Recursive Relations

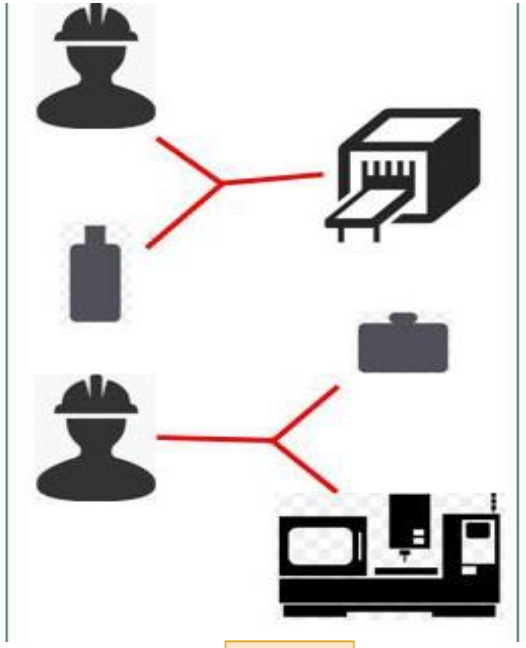


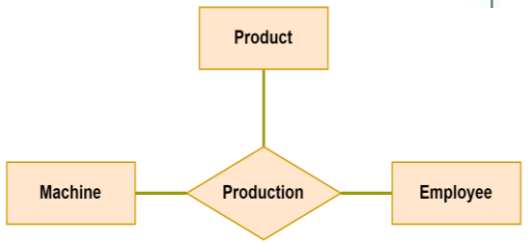


Binary Relation



Ternary Relation



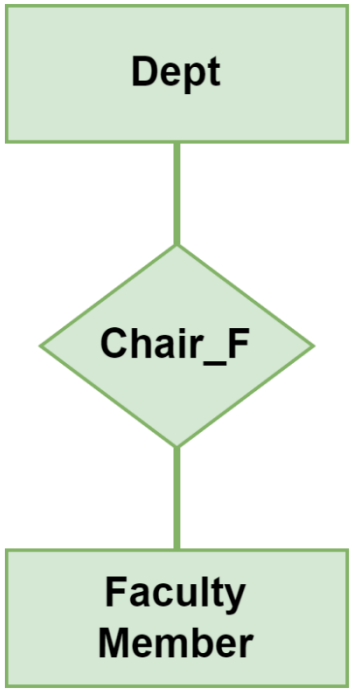


n-n Relation

Employee
Product
Machine
Shift
....

Relationships

- Associates one or more sets of entities.



- All departments have a faculty member who serves as the chair.
- A faculty member can only chair one department.

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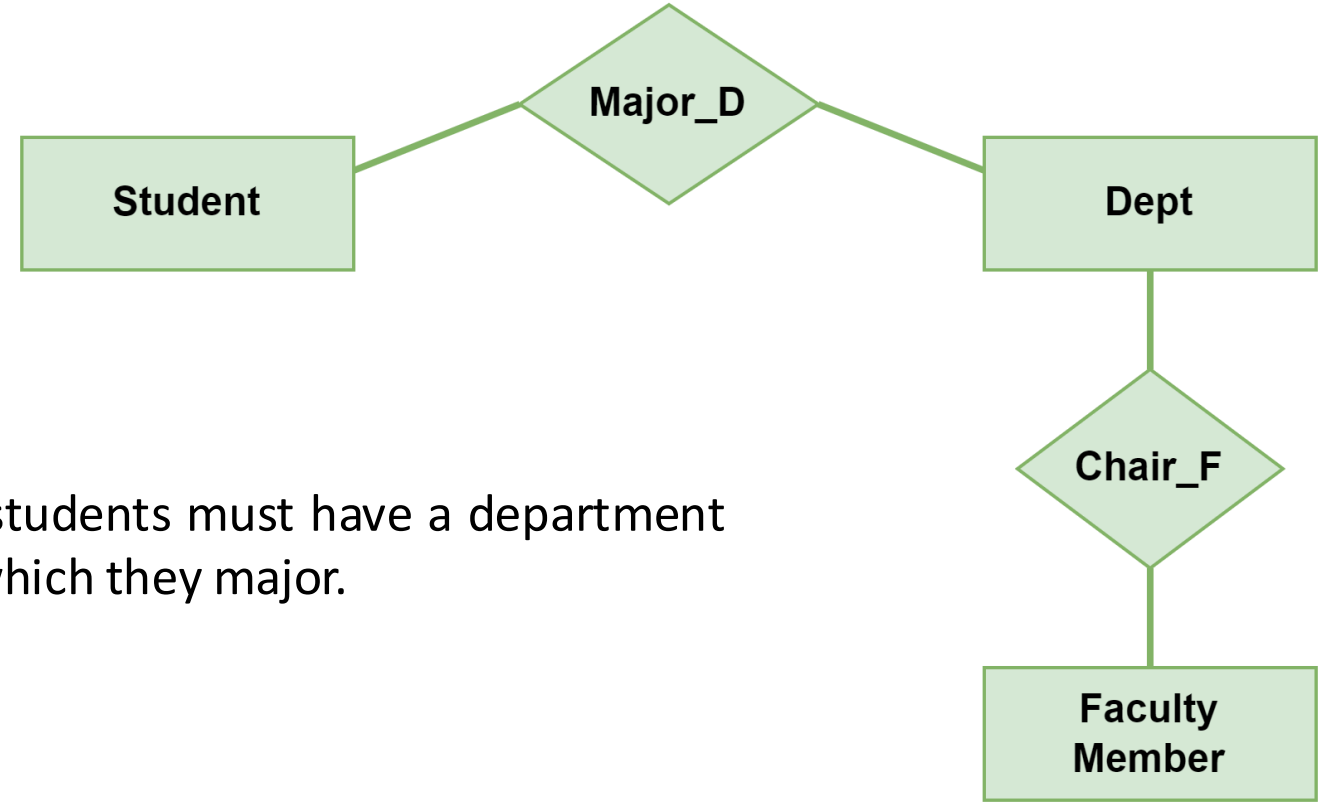

- All students must have a department in which they major.

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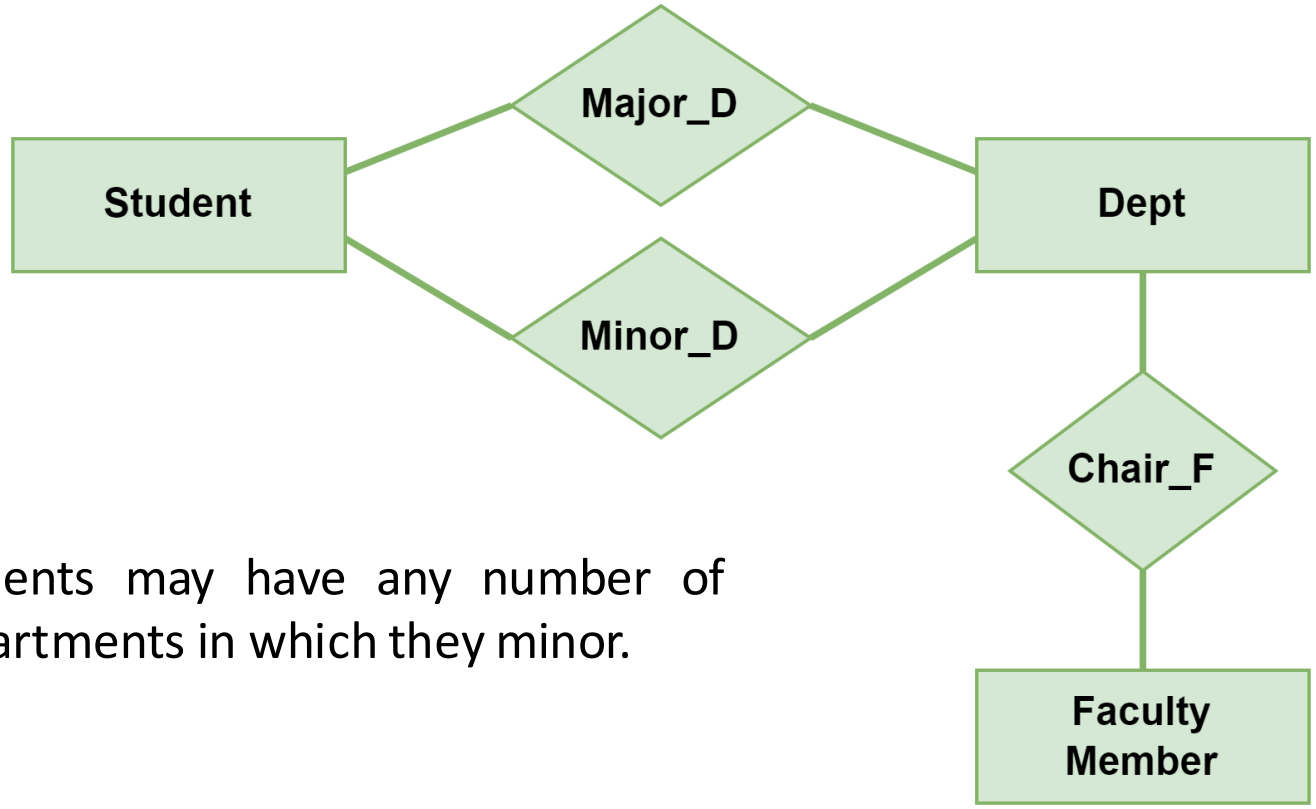

- Students may have any number of departments in which they minor.

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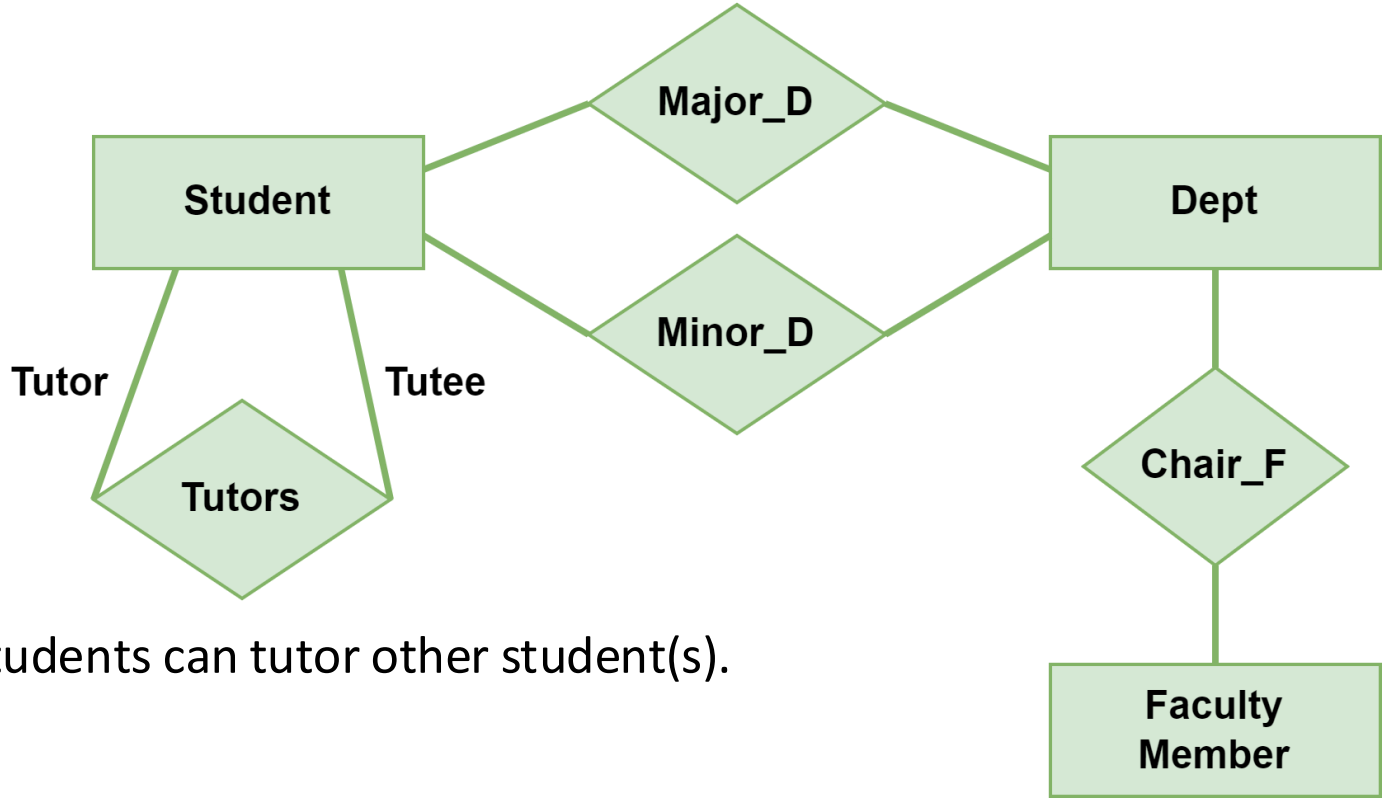
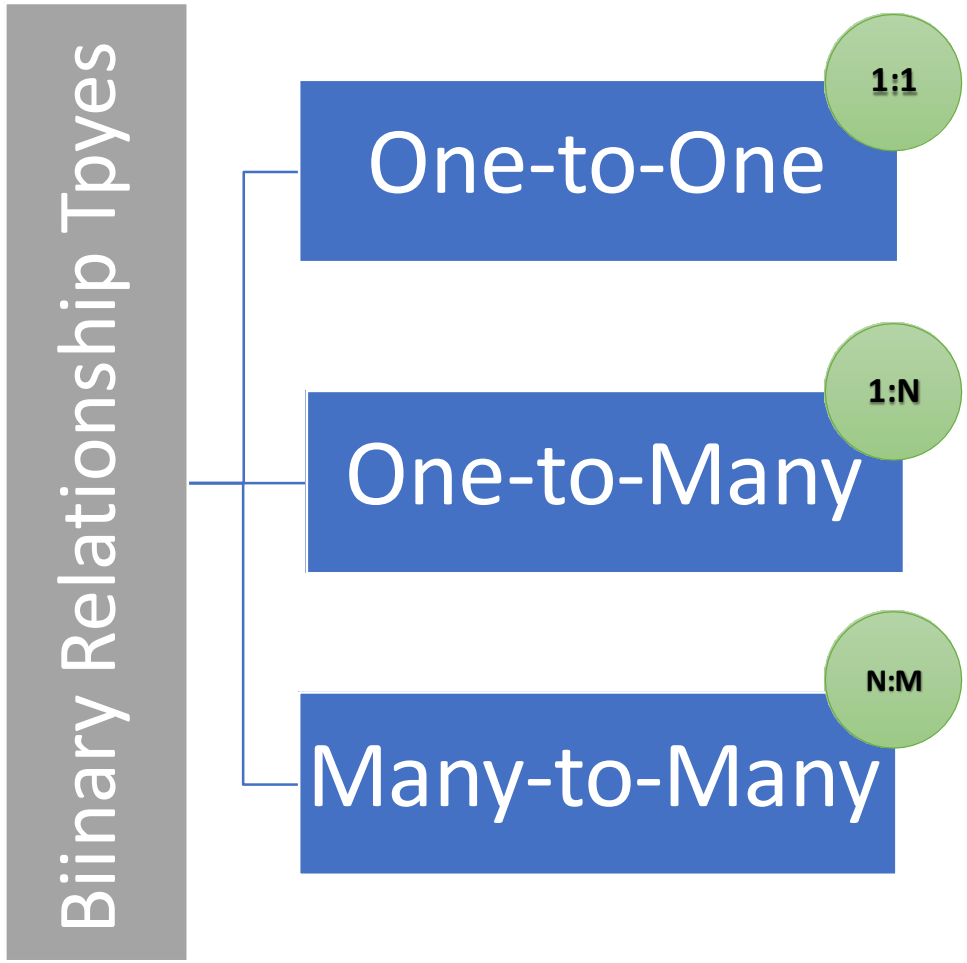
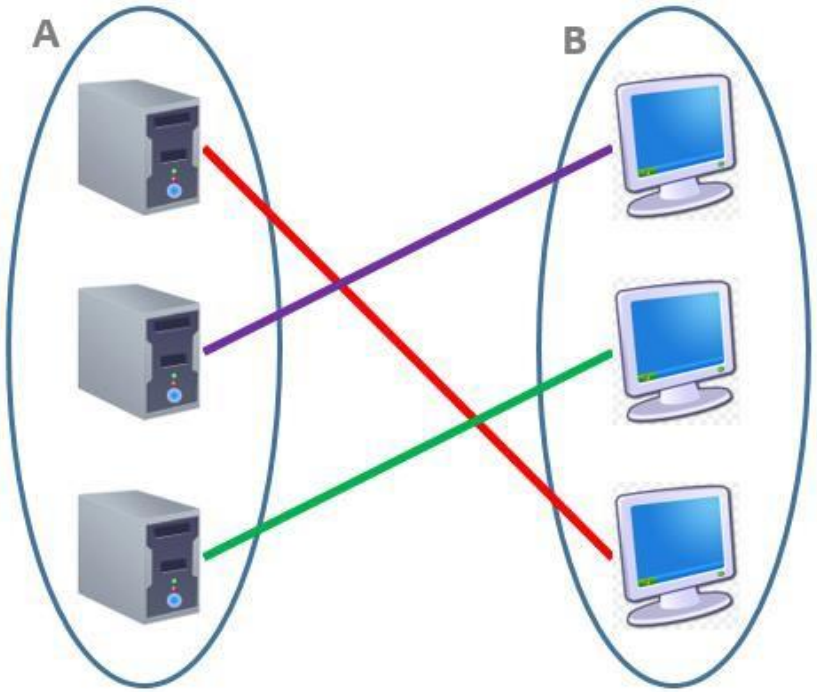


- Students can tutor other student(s).

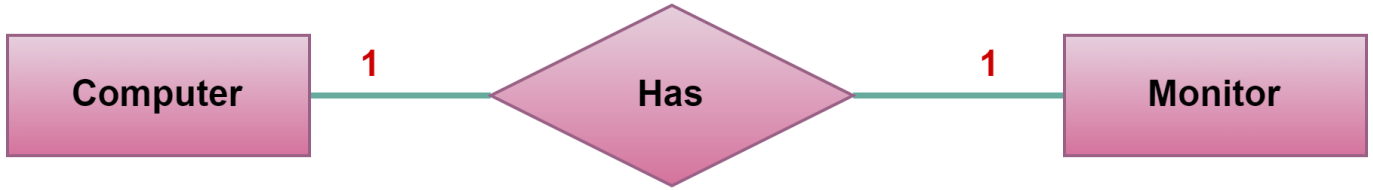
Binary Relationship



One-to-One Relationship



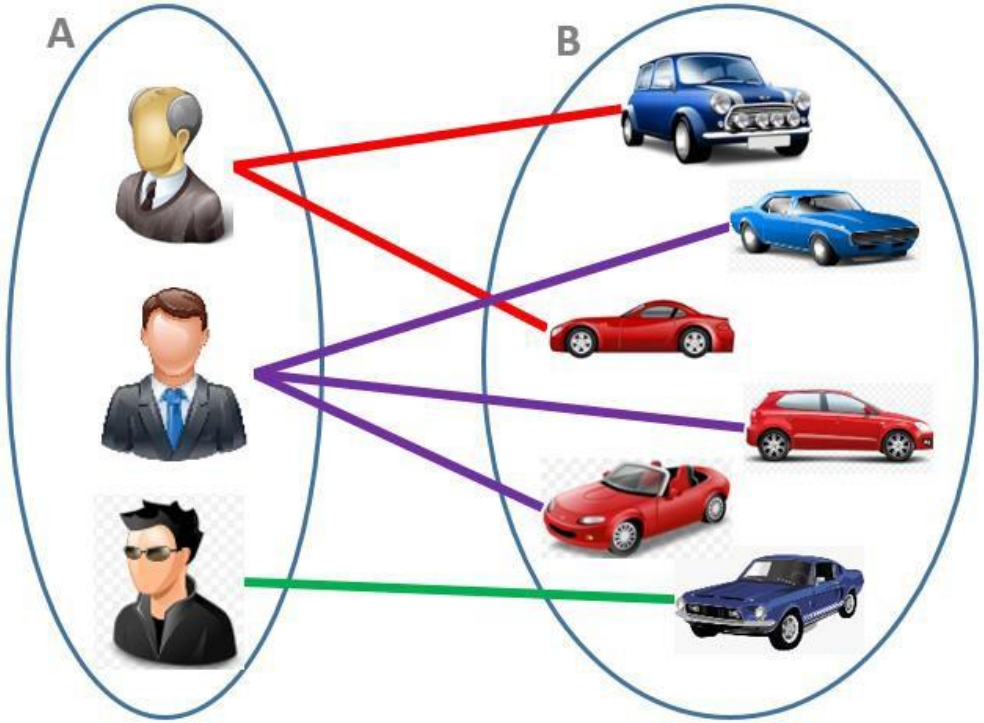
- Each entity in A is associated with at most one entity in B.
- Each entity in B is associated with at most one entity in A.



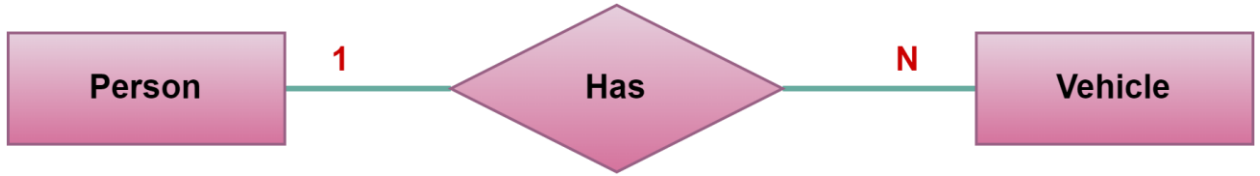
One-to-One Relationship

Entity-1	Entity-2	Description
Student	Class	Each student can be the class representative of at most one class. Each class can have a maximum of one pclass representative.
Computer	Monitor	Each computer can use at most one monitor. Each monitor can be used on a maximum of one computer.

One-to-Many Relationship



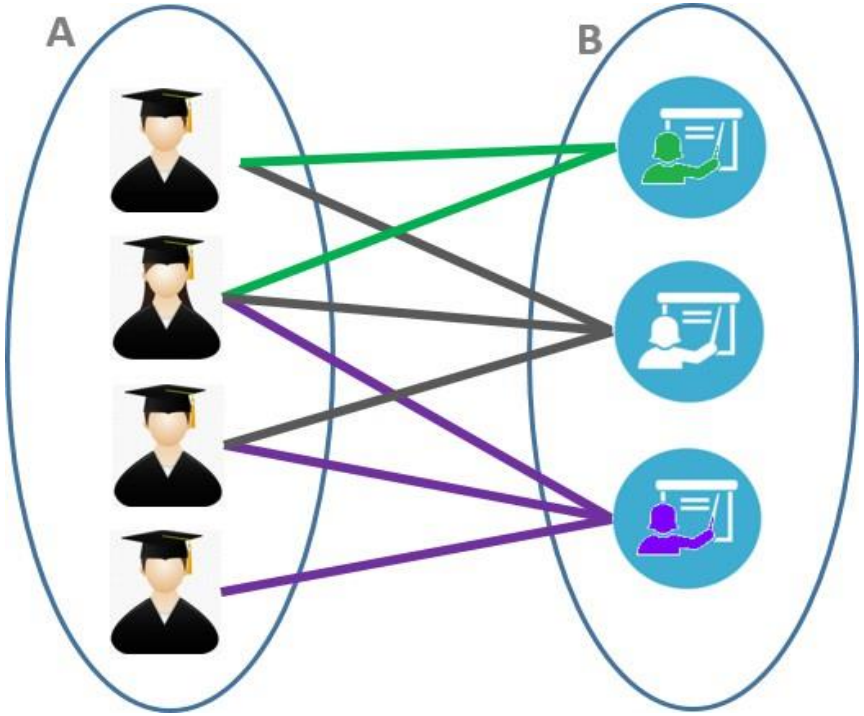
- Each entity in A is associated with more than one entity in B.
- Every entity in B is associated with at most one entity in A.



One-to-Many Relationship

Entity-1	Entity-2	Description
Student	Class	Each student can be in a maximum of one class. Each class can contain more than one student.
Company	Computer	Each company can have more than one computer. Each computer can belong to at most one company.

Many-to-Many Relationship



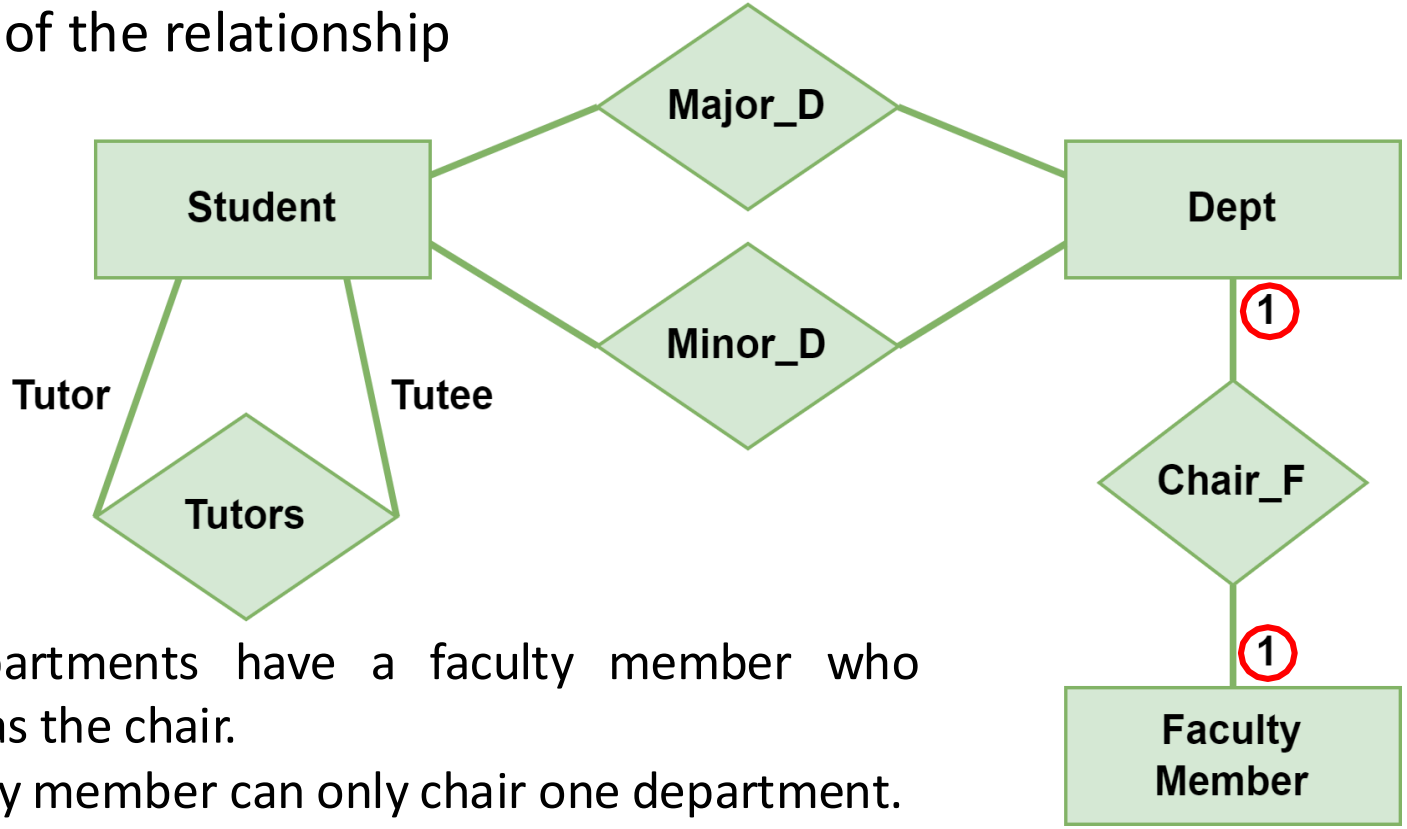
- Each entity in A can be associated with more than one entity in B.
- The same condition is valid for the entity in B.

Many-to-Many Relationship

Entity-1	Entity-2	Description
Student	Course	Each student can take more than one course. Each course can be taken by more than one student.
Company	Customer	Each company can sell products to more than one customer. Each customer can also purchase products from more than one company.

Cardinality Ratios

- Constrains the number of entities that can participate in each role of the relationship



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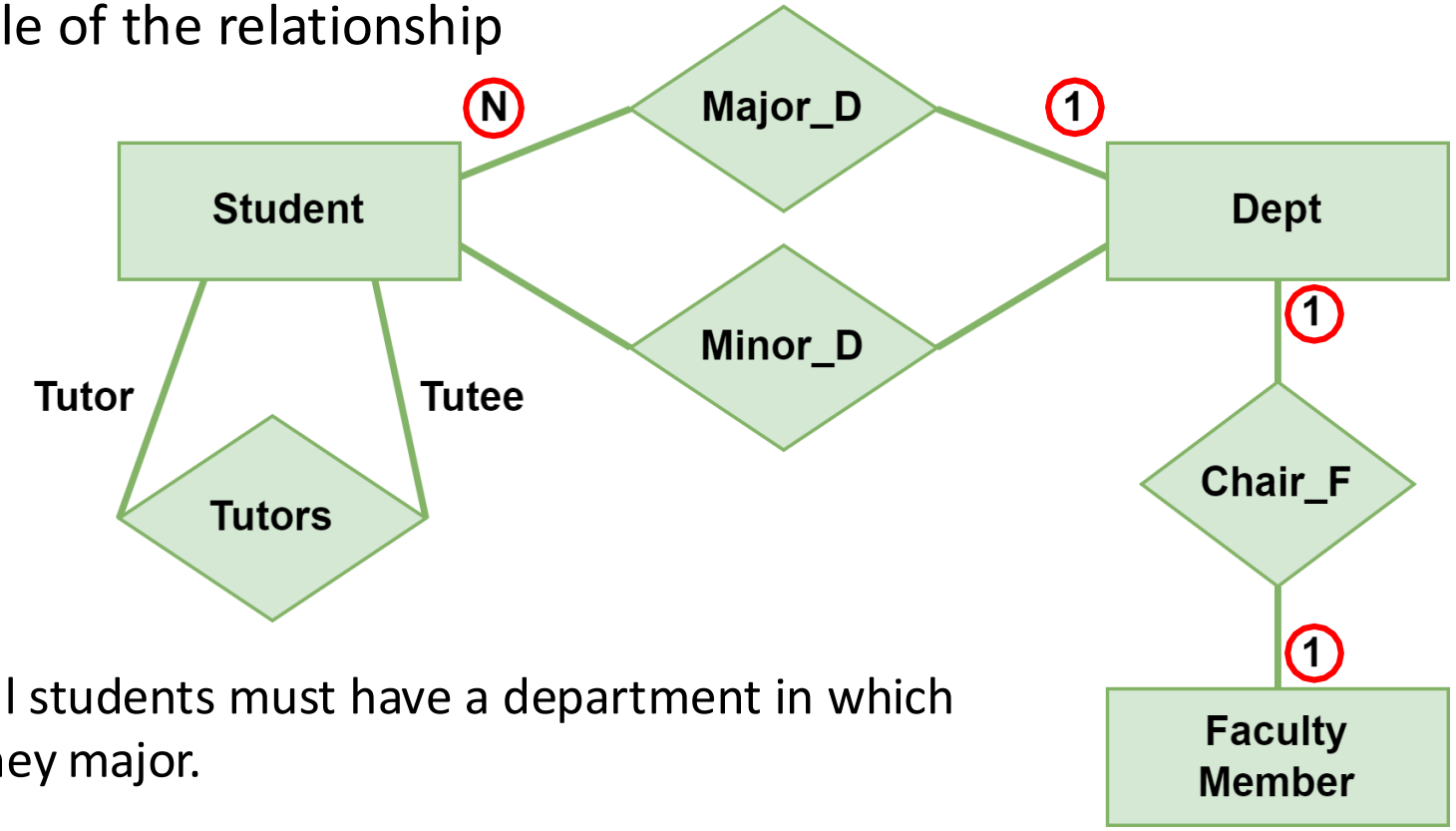

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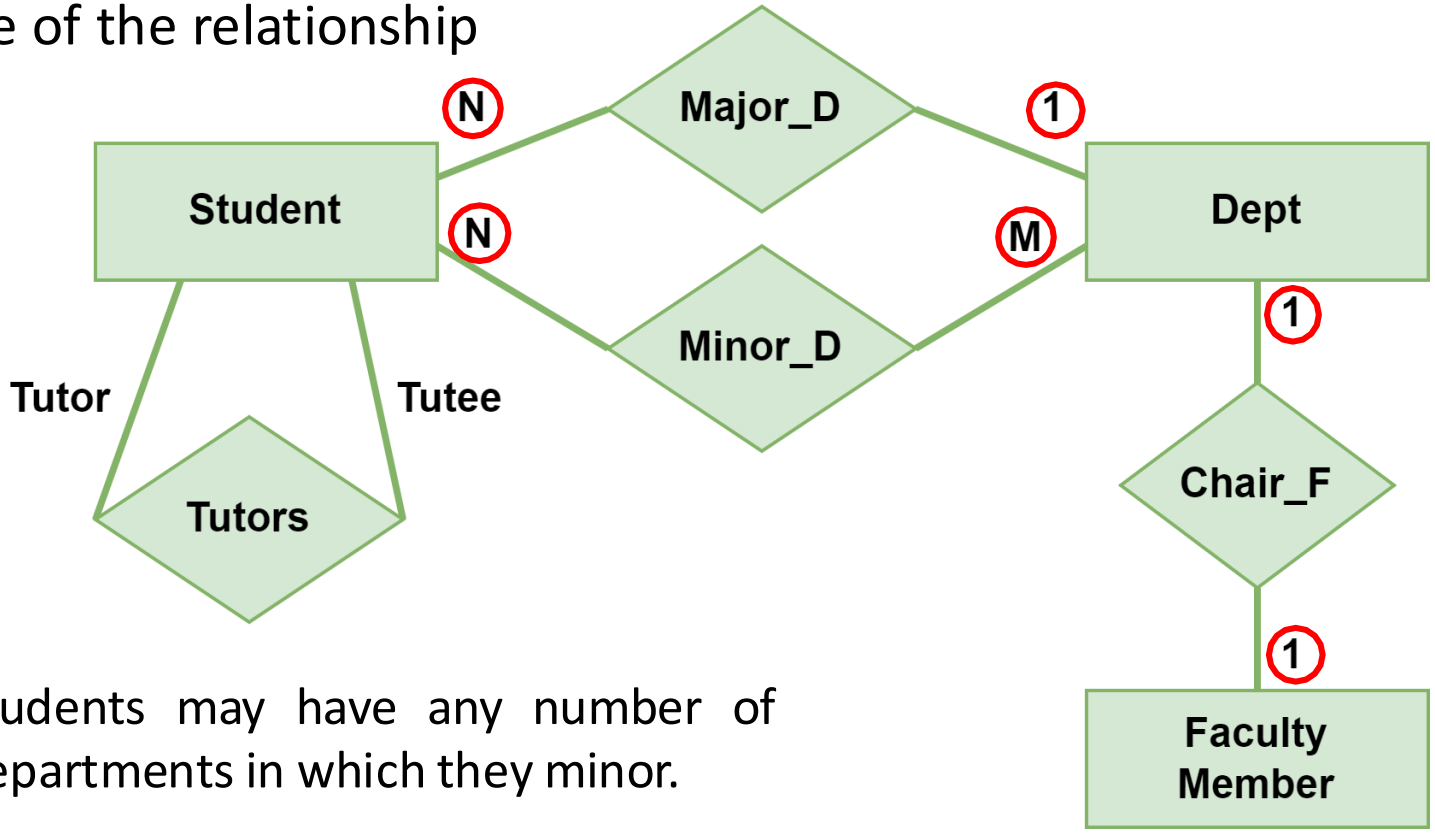

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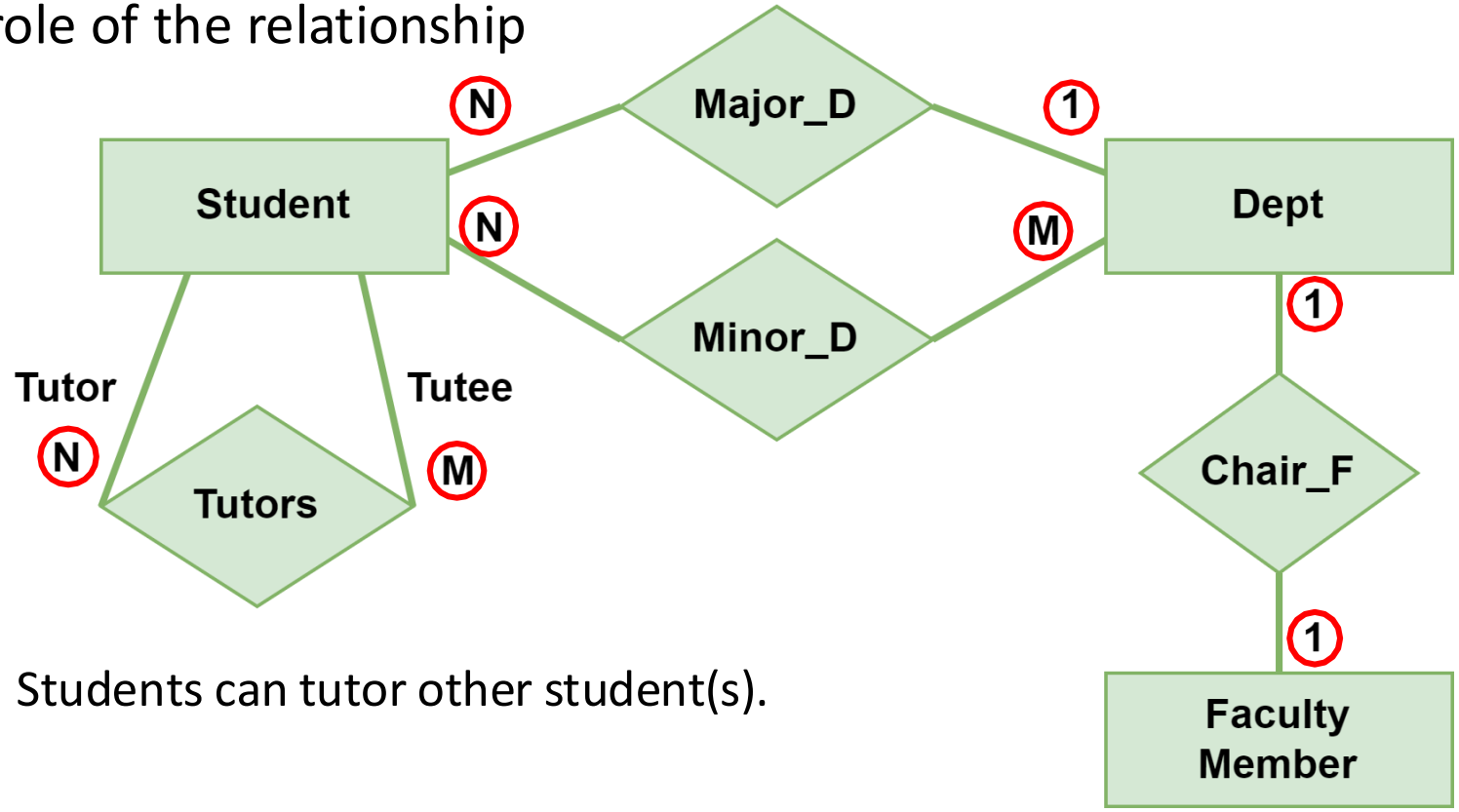
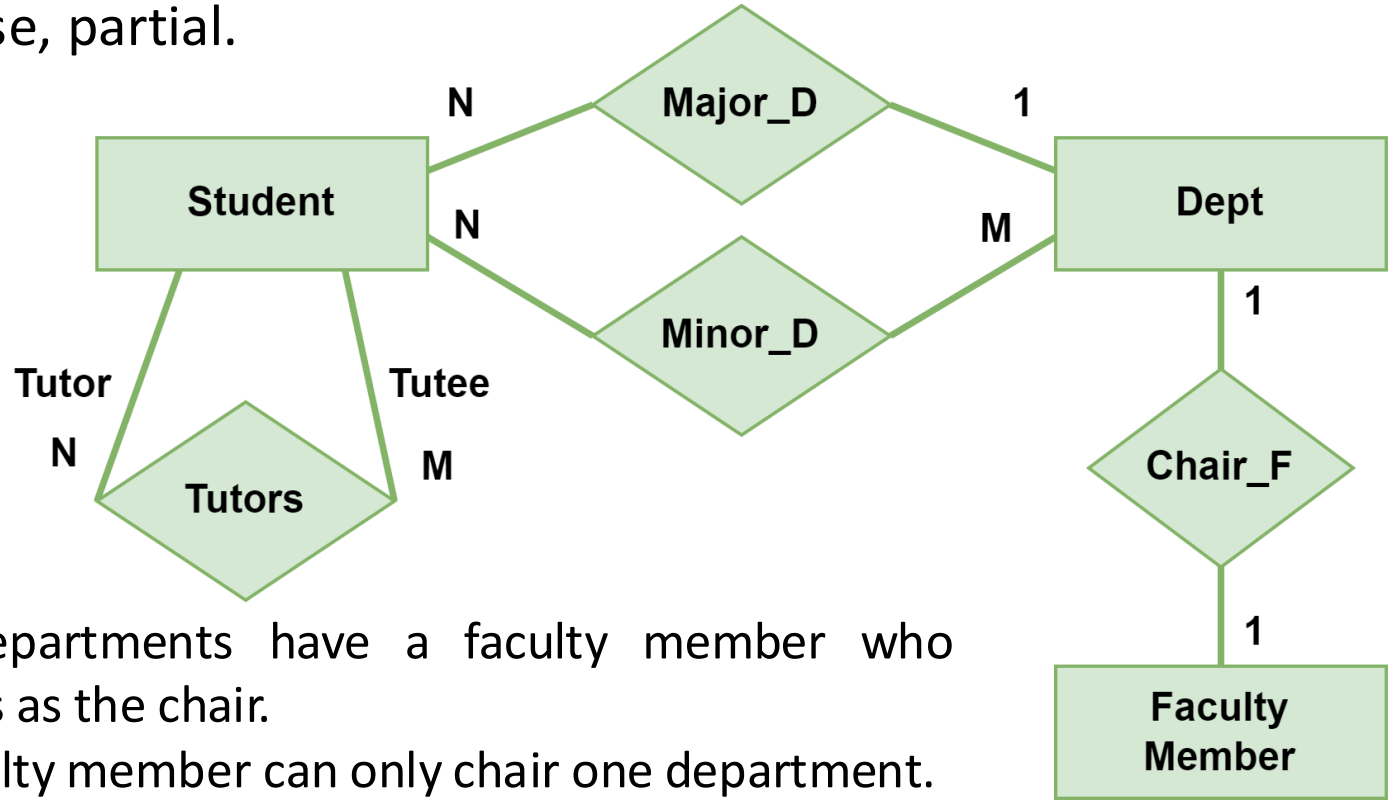


- Students can tutor other student(s).

Structural Constraints

- If an entity does not exist unless it appears with an entity in a relationship, the participation is total (existence dependency). Else, partial.



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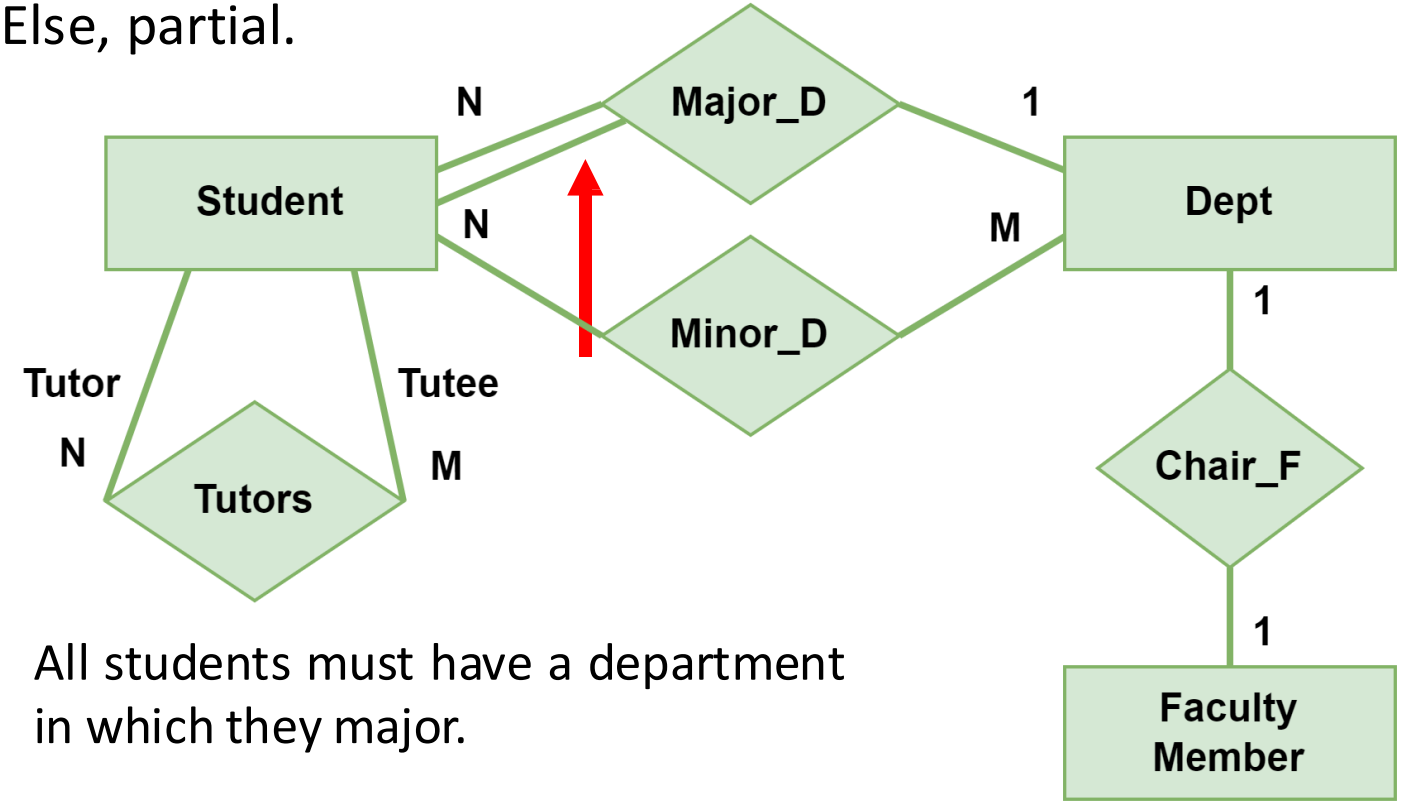

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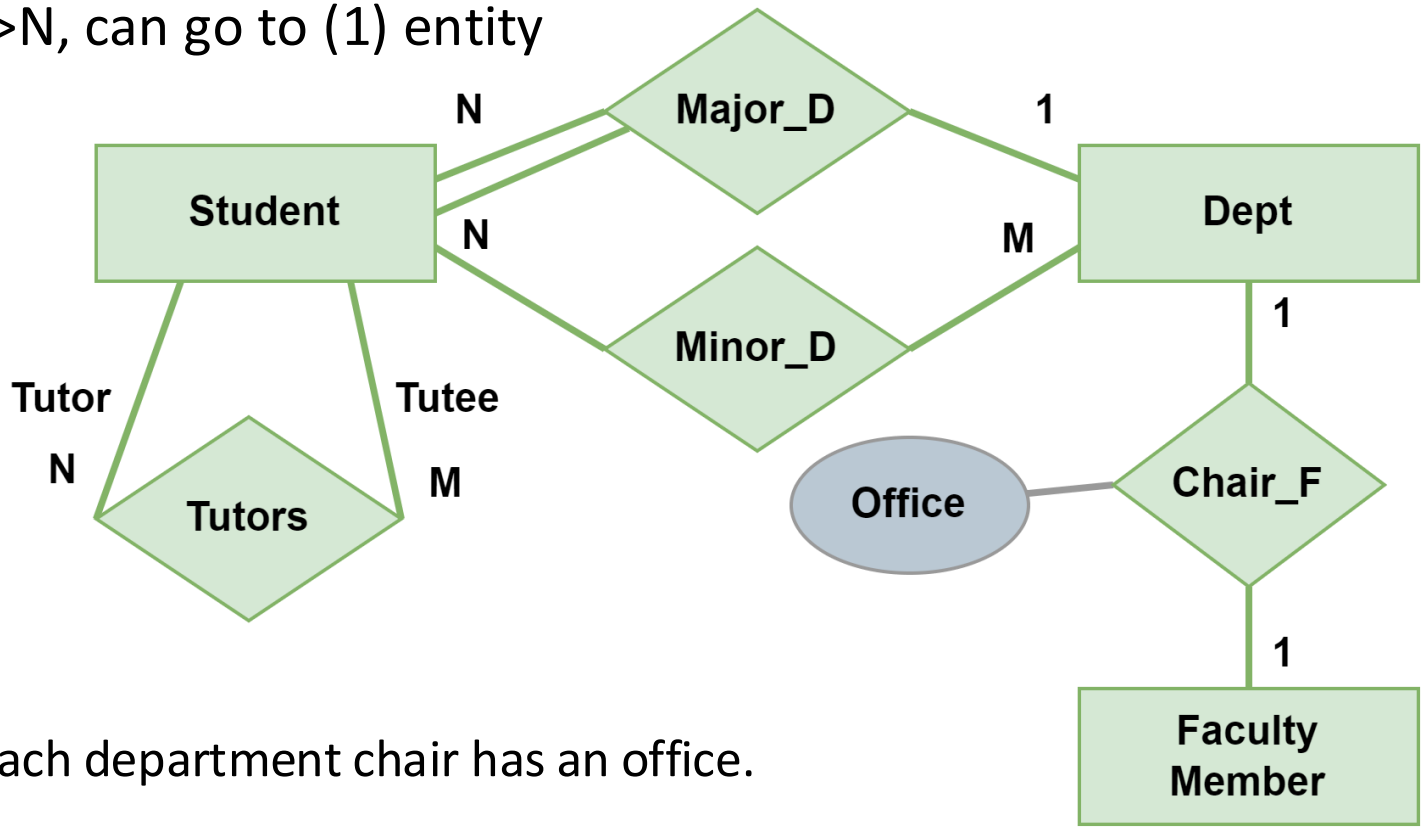
Attributes of Relationships

- 1->1, can go to either entity
- 1->N, can go to (1) entity



- Each department chair has an office.

Attributes of Relationships

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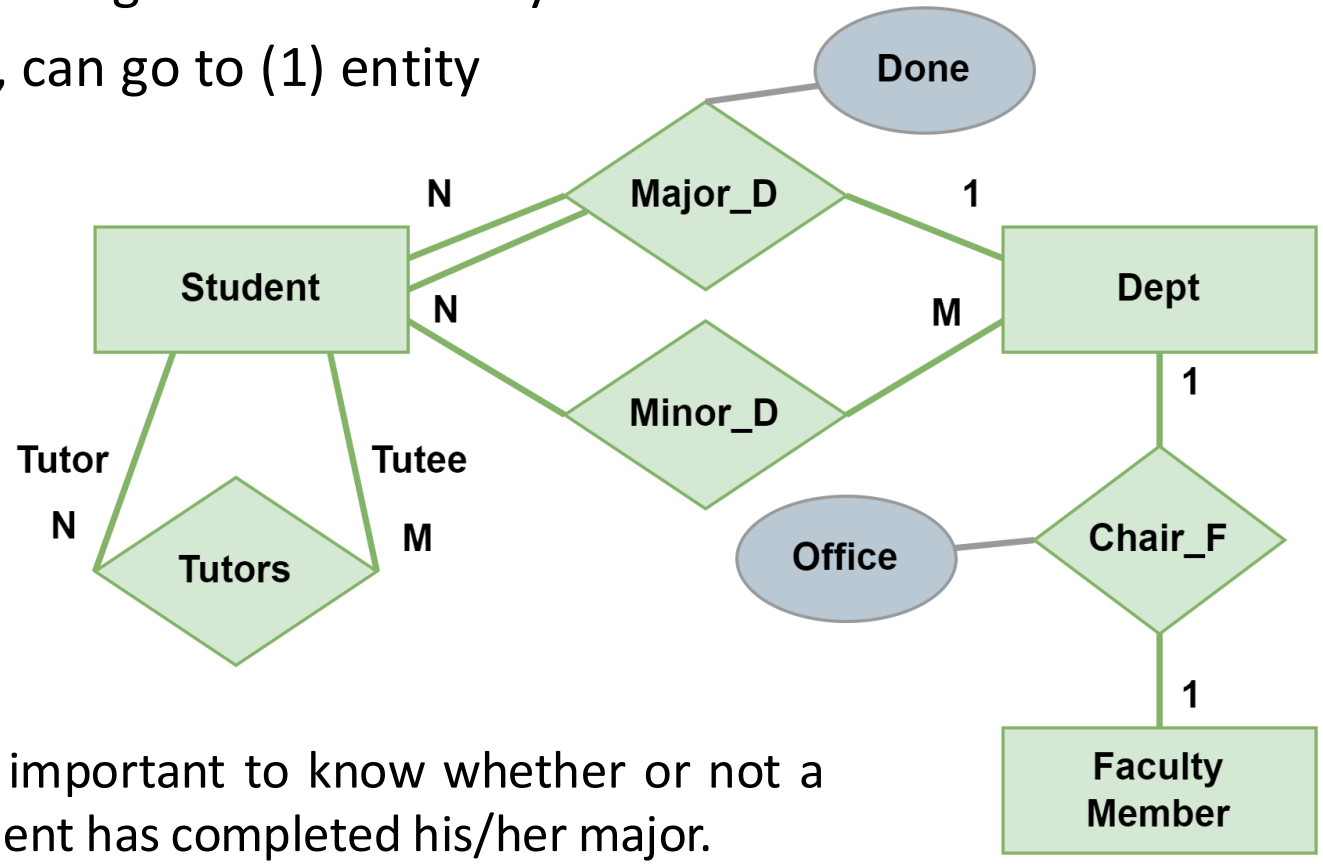
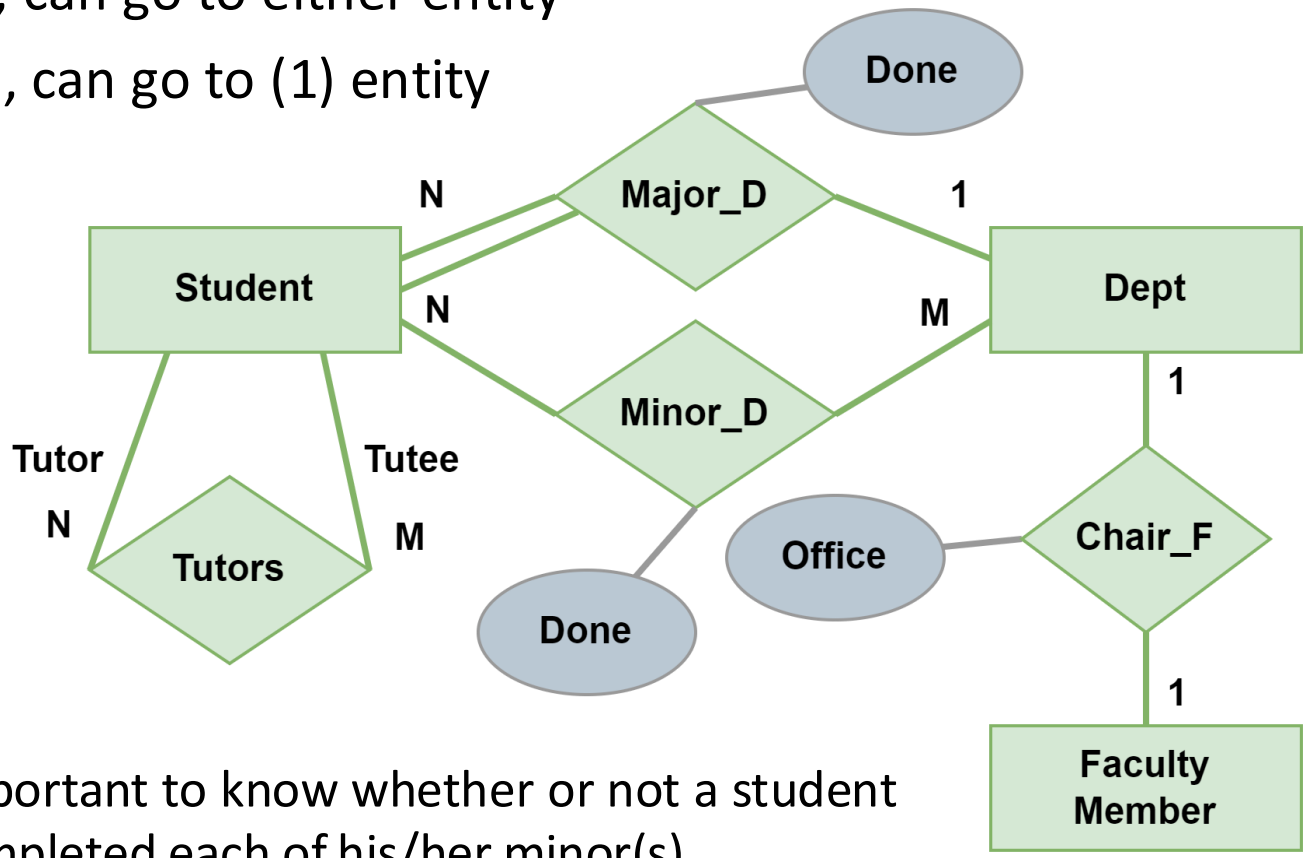
- 1->N, can go to (1) entity



- It is important to know whether or not a student has completed his/her major.

Attributes of Relationships

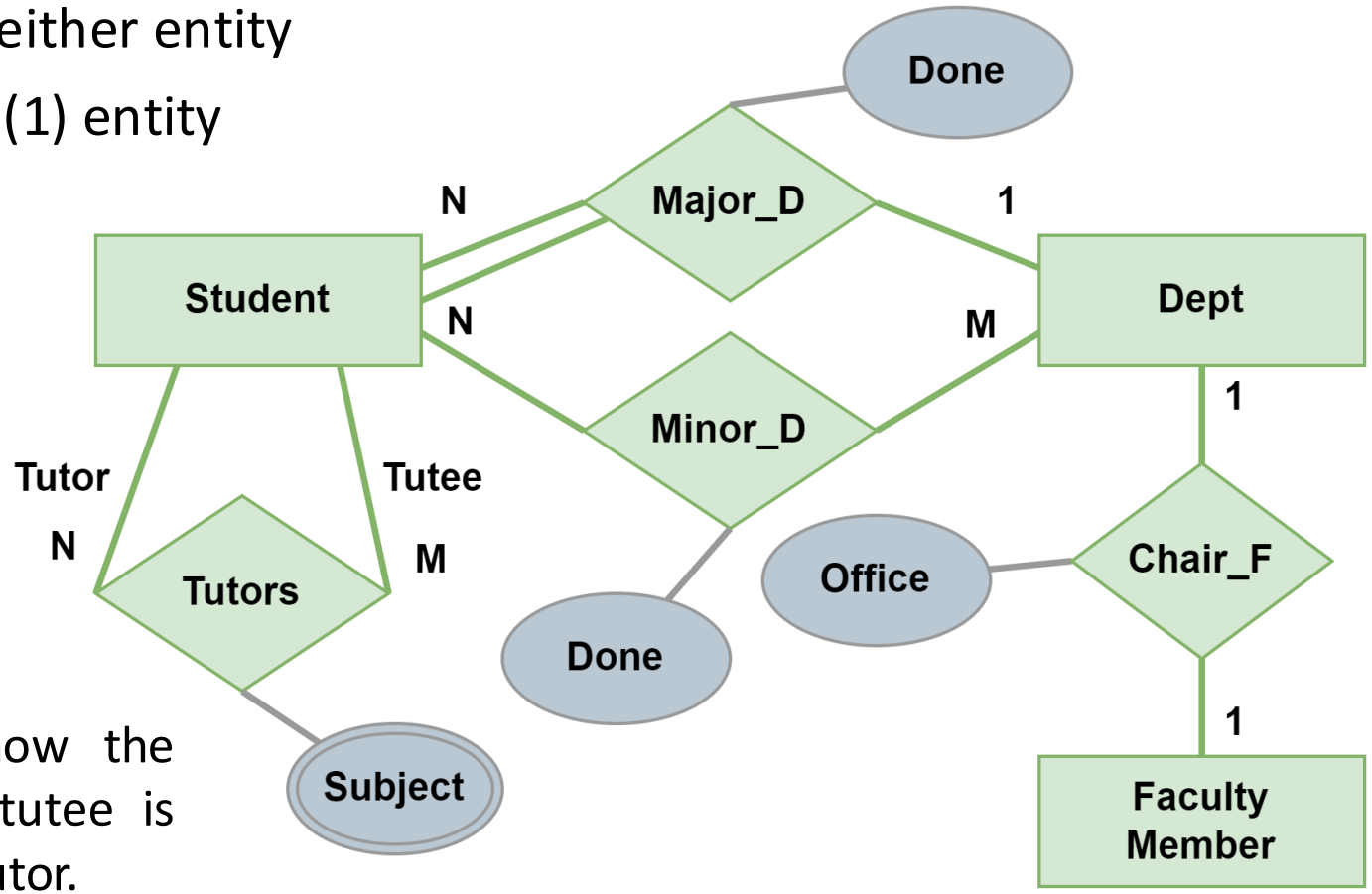
- 1->1, can go to either entity
- 1->N, can go to (1) entity



- It is important to know whether or not a student has completed each of his/her minor(s).

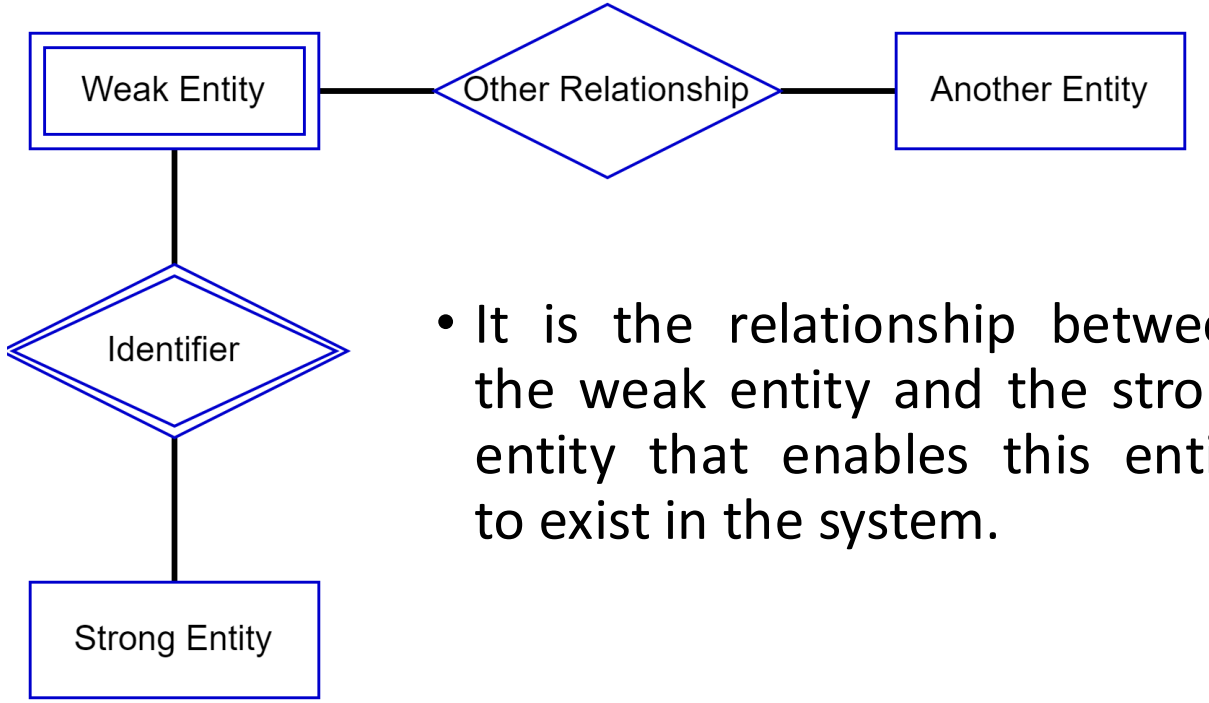
Attributes of Relationships

- 1->1, can go to either entity
- 1->N, can go to (1) entity



- It is important to know the subject(s) in which a tutee is being tutored by each tutor.

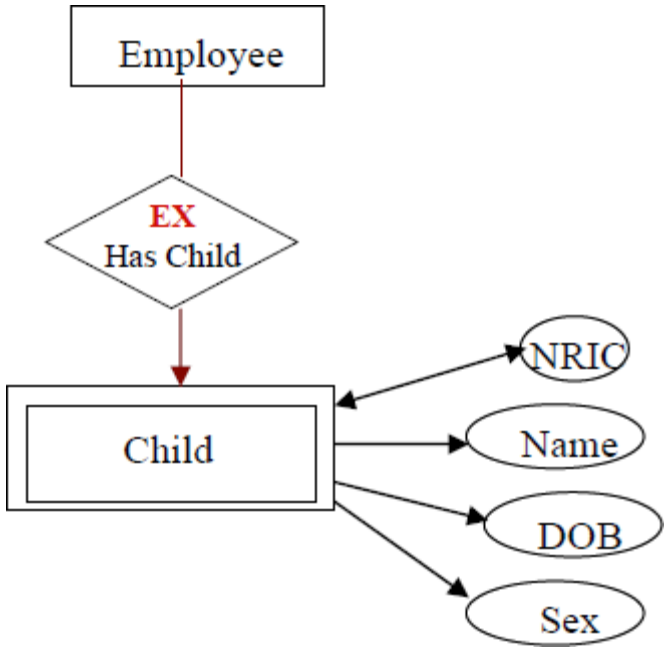
Identifying Relationships



- It is the relationship between the weak entity and the strong entity that enables this entity to exist in the system.

Existence-dependency (EX)

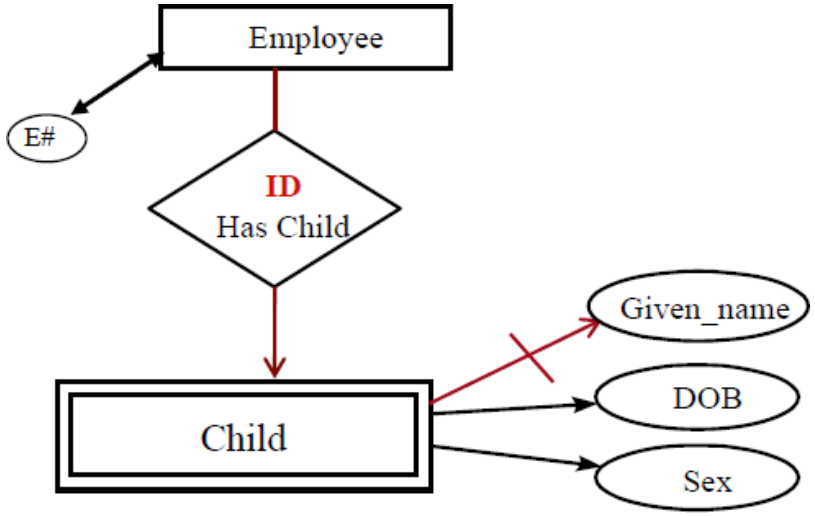
- The existence of a Child entity depends on the existence of an associated EMPLOYEE entity. Thus, if an Employee entity is deleted, its associated Child entities are also deleted.
- The dependent entity type Child is a weak entity type (represented by a double rectangle), and Employee is a regular entity type.
- Relationship type which involves a weak entity type is called existence- dependency relationship type (denoted with “EX” together with a relationship type name).



Note: An EX relationship type is a **1:m (one to many)** relationship type.

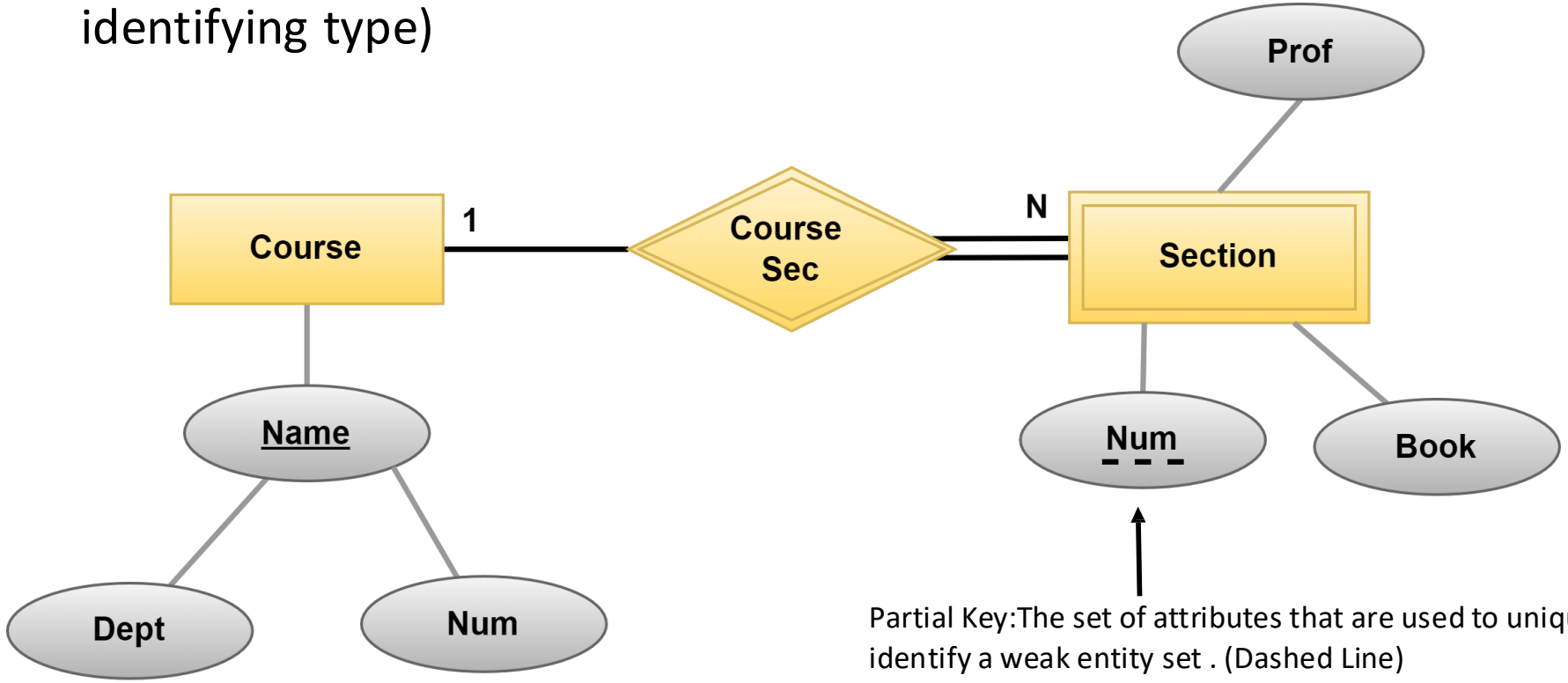
Identifier-dependency (ID)

- An entity cannot be identified by the value of its own attributes (**does not have a candidate key**), but has to be identified by its relationship with other entity. Such a relationship is called identifier-dependency relationship.
- Note: The line on the arrow to the attribute Given_name indicates this attribute together with the identifier of Employee (i.e. E#) form the identifier of the weak entity type Child.
- Note: By the original definitions, an identifier-dependency relationship type (denoted by **ID**) is also an existence-dependency relationship type. However, we should not just indicate an **ID** as **EX**.



Weak Entities

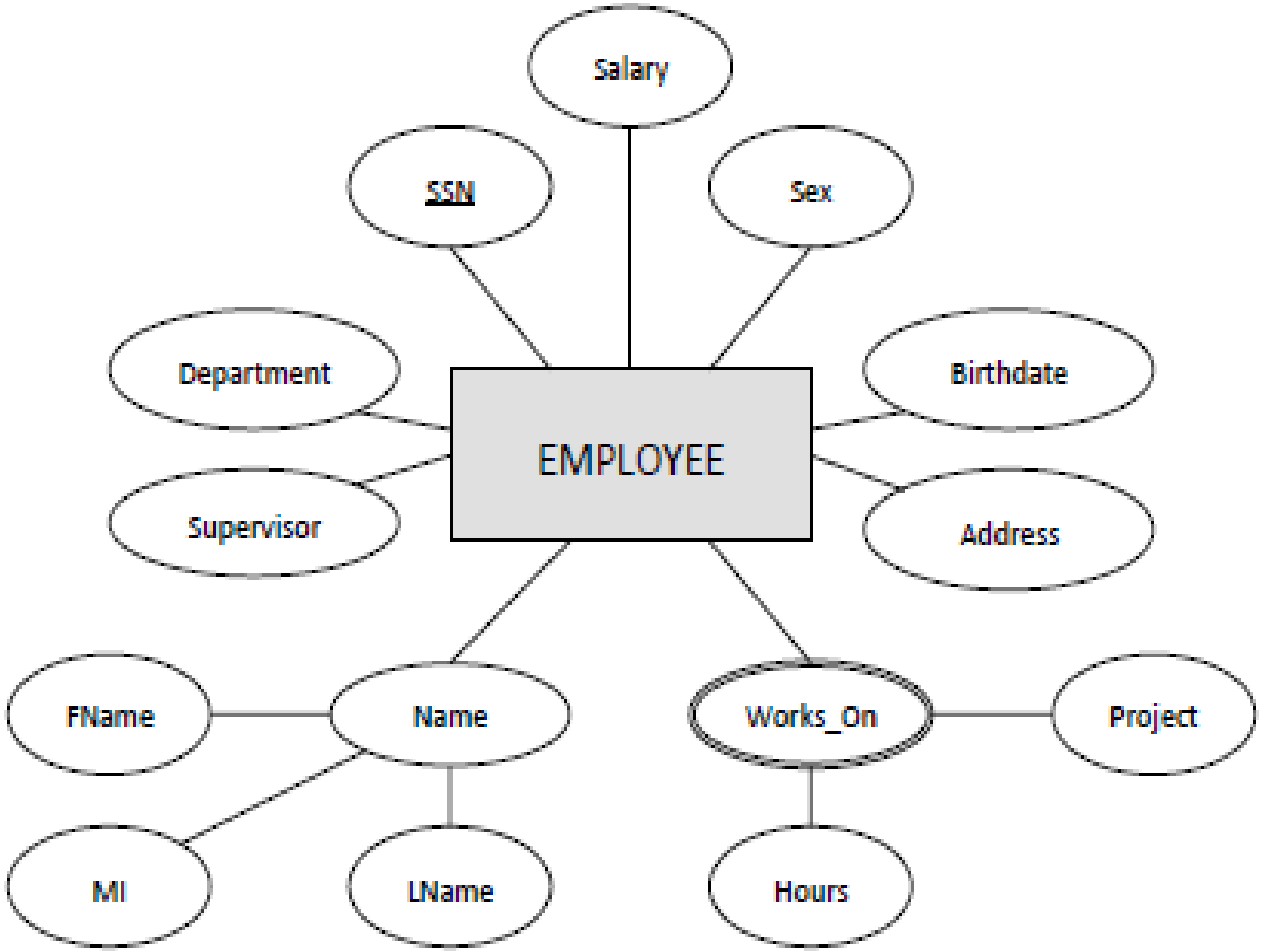
- Entity types that do not have key attributes of their own are weak; instead identified by relation to specific entity of another type (the identifying type)



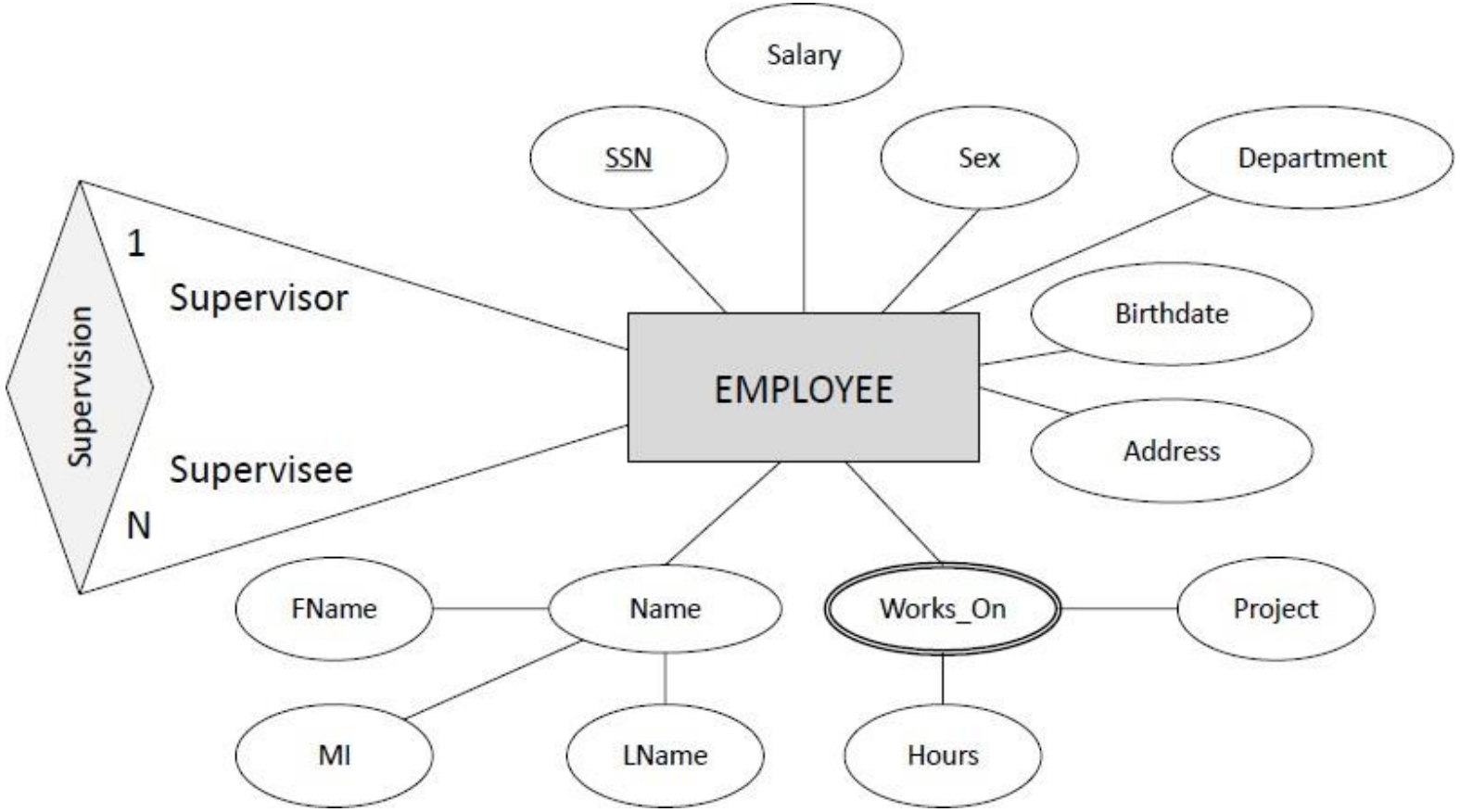
Partial Key:The set of attributes that are used to uniquely identify a weak entity set . (Dashed Line)

Revise the example

- We store each employee's name (first, last, MI), Social Security number (SSN), Street address, salary, sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).

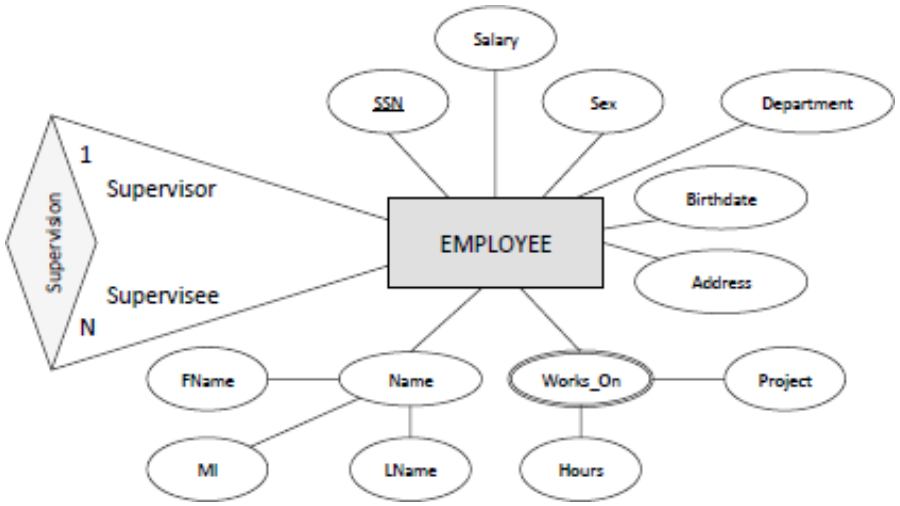


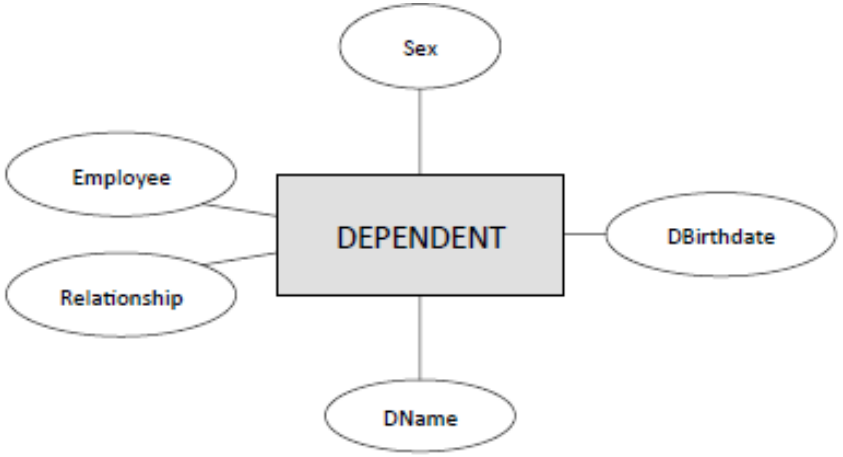
Revise the example



Revise the example

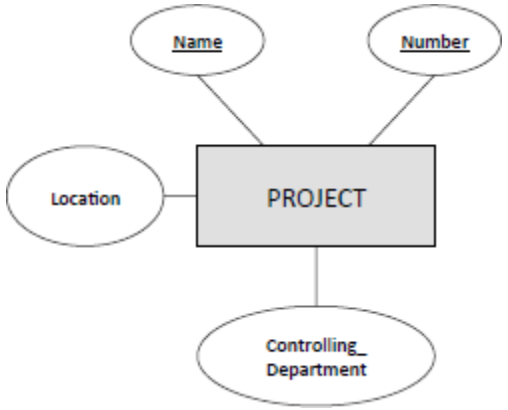
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, sex, birth date, and relationship to the employee.

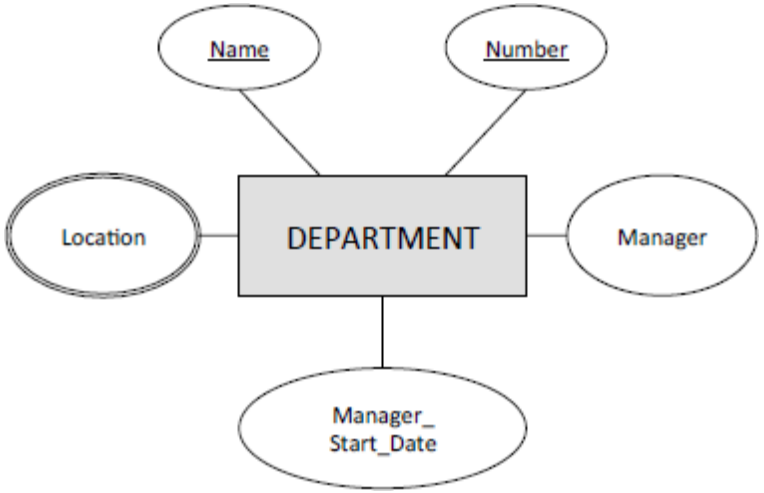




Revise the example

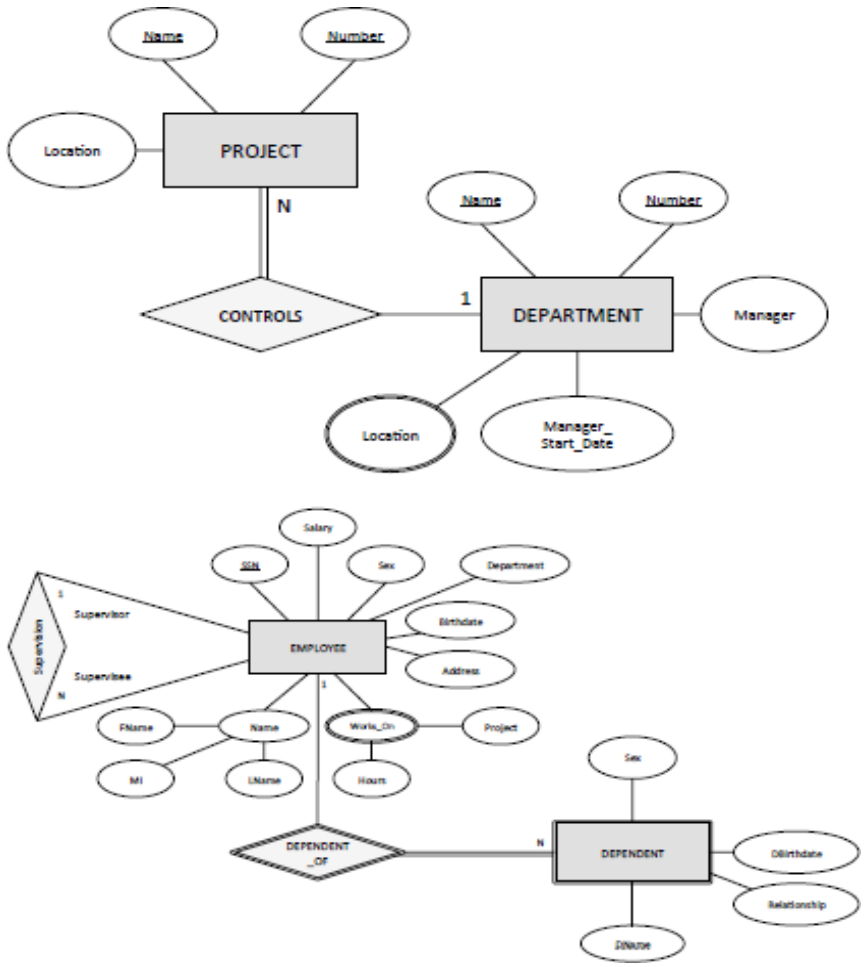
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.





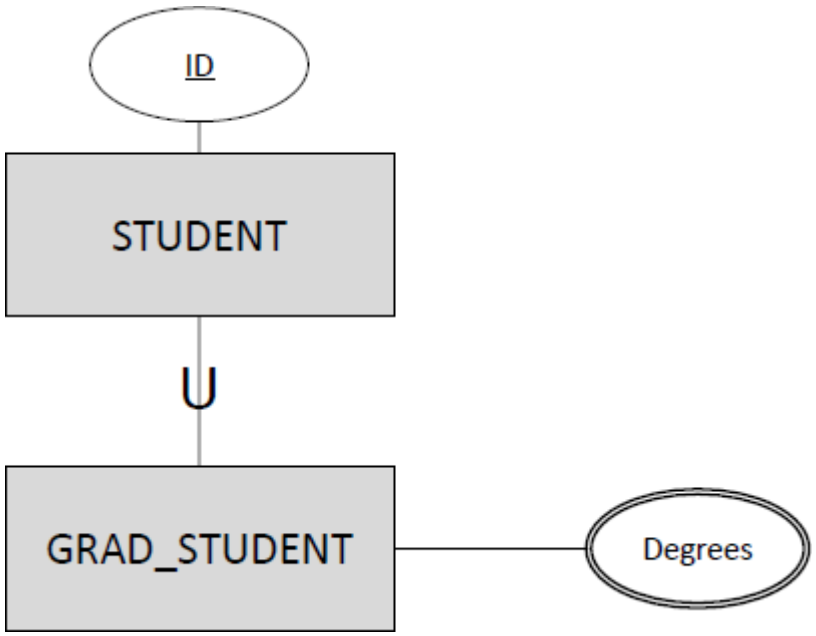
Revise the example

- Each department has a particular employee who manages the department.
- An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project.



Specialization/Generalization

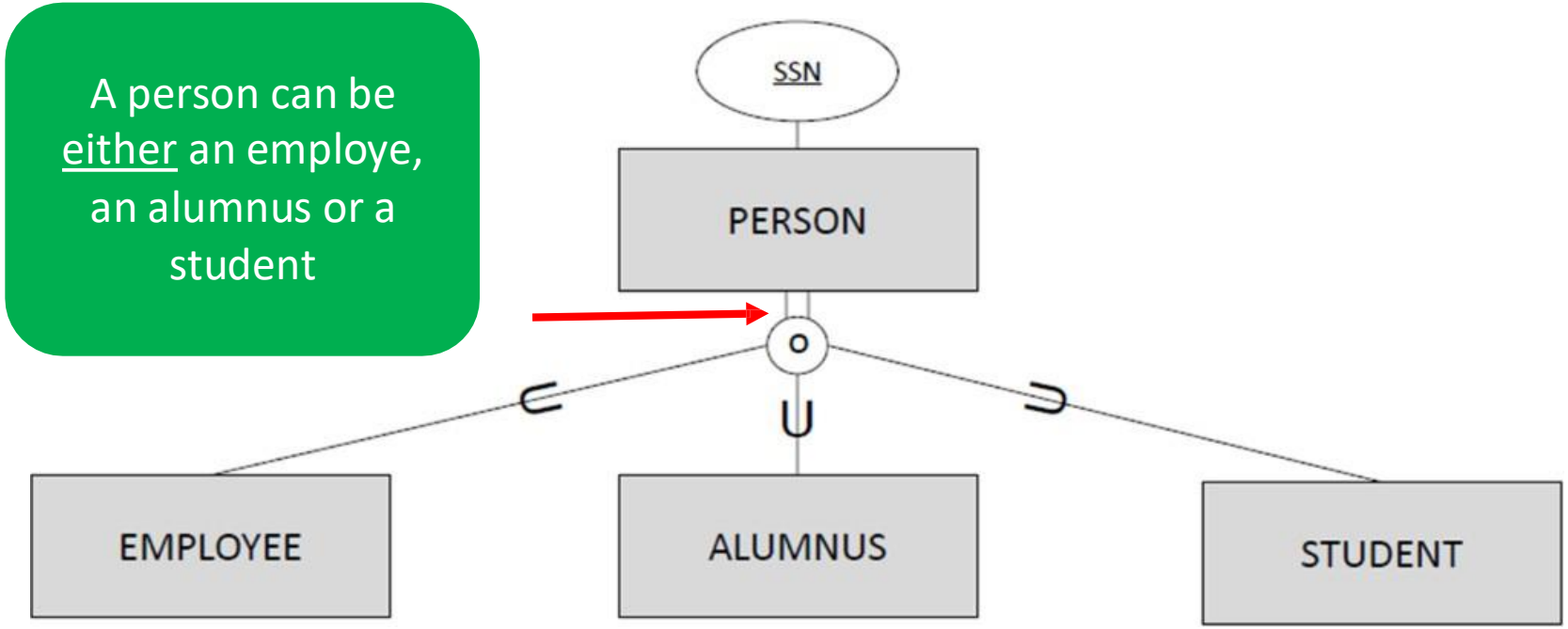
- Only a subset of entities within a type have certain attributes or participate in certain relationships



Multiple Subtypes: Overlapness

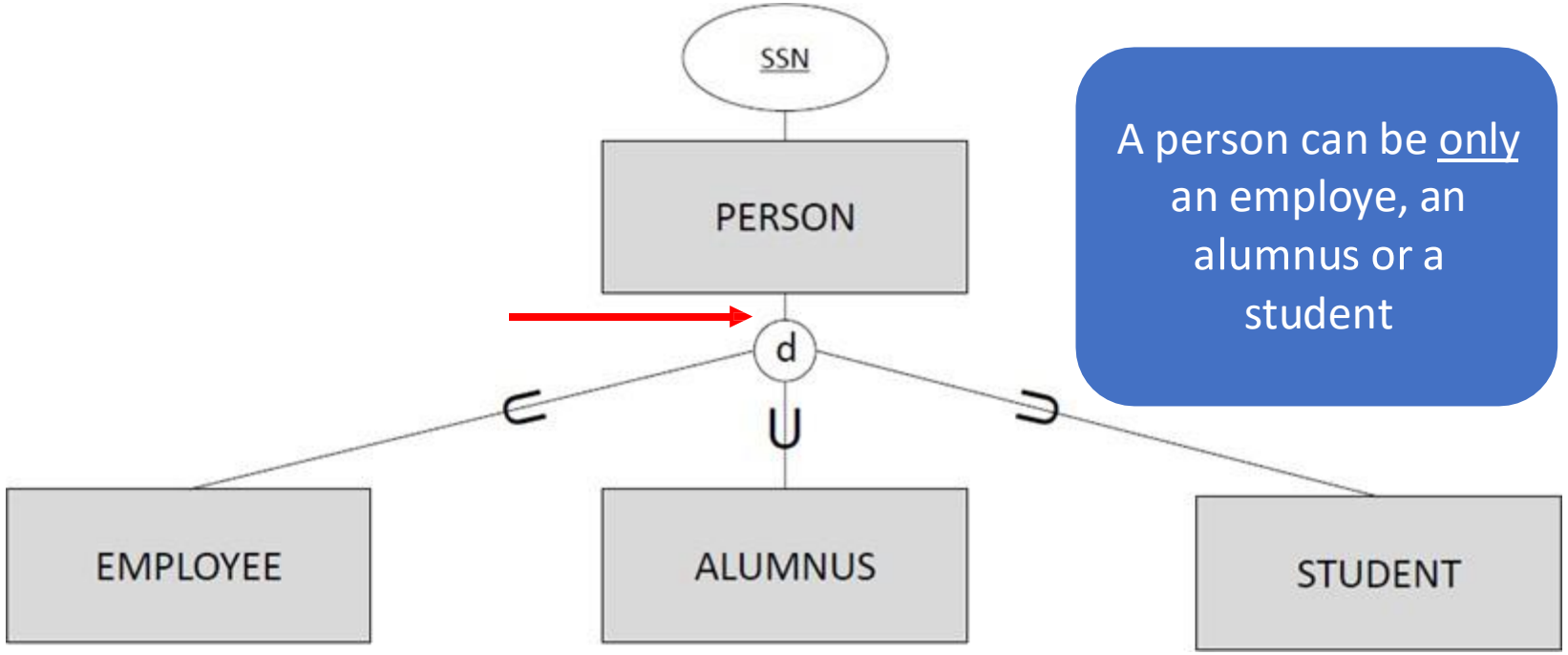
- (o)verlap: may be more than one
- (d)isjoint: entities may only be one subtype

A person can be <u>either</u> an employe, an alumnus or a student



Multiple Subtypes: Disjointedness

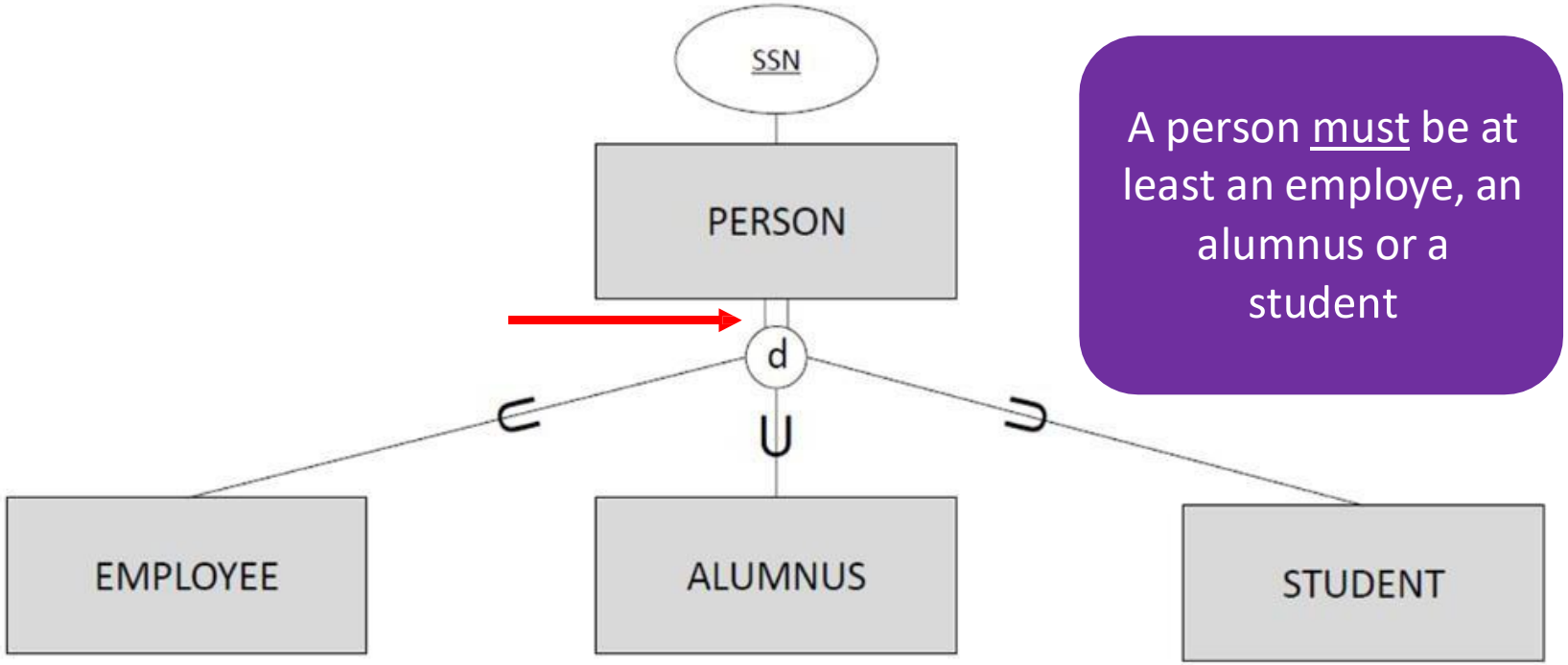
- (o)verlap: may be more than one
- (d)isjoint: entities may only be one subtype



A person can be <u>only</u> an employe, an alumnus or a student

Multiple Subtypes: Completeness

- Similar to relationships; can be total (must belong to subtypes) or partial (can belong)



A person must be at least an employe, an alumnus or a student

English Sentence Structure and ER Diagram

- In order to construct a database using an ER diagram, the database designer not only has to interview users but also must study the system specification documents which are written in some natural language, such as English.
- Some guidelines/rules for translating English sentences into ER diagrams are presented below:

Guideline 1: A **common noun** (such as student and employee) in English corresponds to an entity type in an ER diagram:

common noun → entity type

Note: Proper nouns are entities not entity types, e.g. John, Singapore, New York City.

English Sentence Structure and ER Diagram

Guideline 2: A **transitive verb** in English corresponds to a relationship type in an ER diagram:

transitive verb → relationship type

Note: A transitive verb must have an object.

E.g. A person may own one or more cars and a car is owned by only one person.

English Sentence Structure and ER Diagram

Guideline 3: An **adjective** in English corresponds to an attribute of an entity type in an ER diagram:

adjective → attribute of entity type

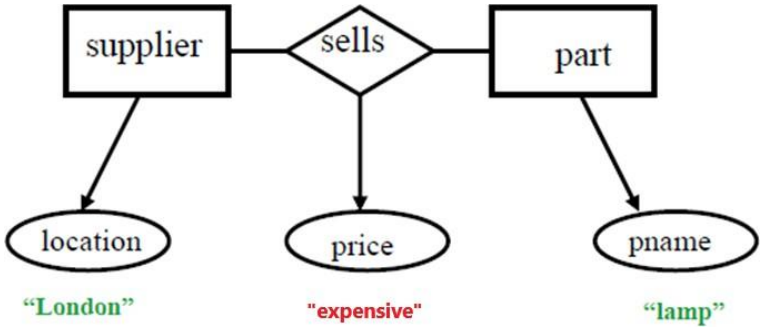
E.g. A London supplier, a red part, a male person.

Guideline 4: An **adverb** in English corresponds to an attribute of a relationship type in an ER diagram:

adverb → attribute of relationship type

E.g. A London supplier sells a part with part name lamp for $50 which is expensive.

London and Lamp are adjectives (and attribute values) of supplier and part, and Expensive is an adverb which is an attribute of the relationship type sells.



Design a conceptual schema by an ER diagram

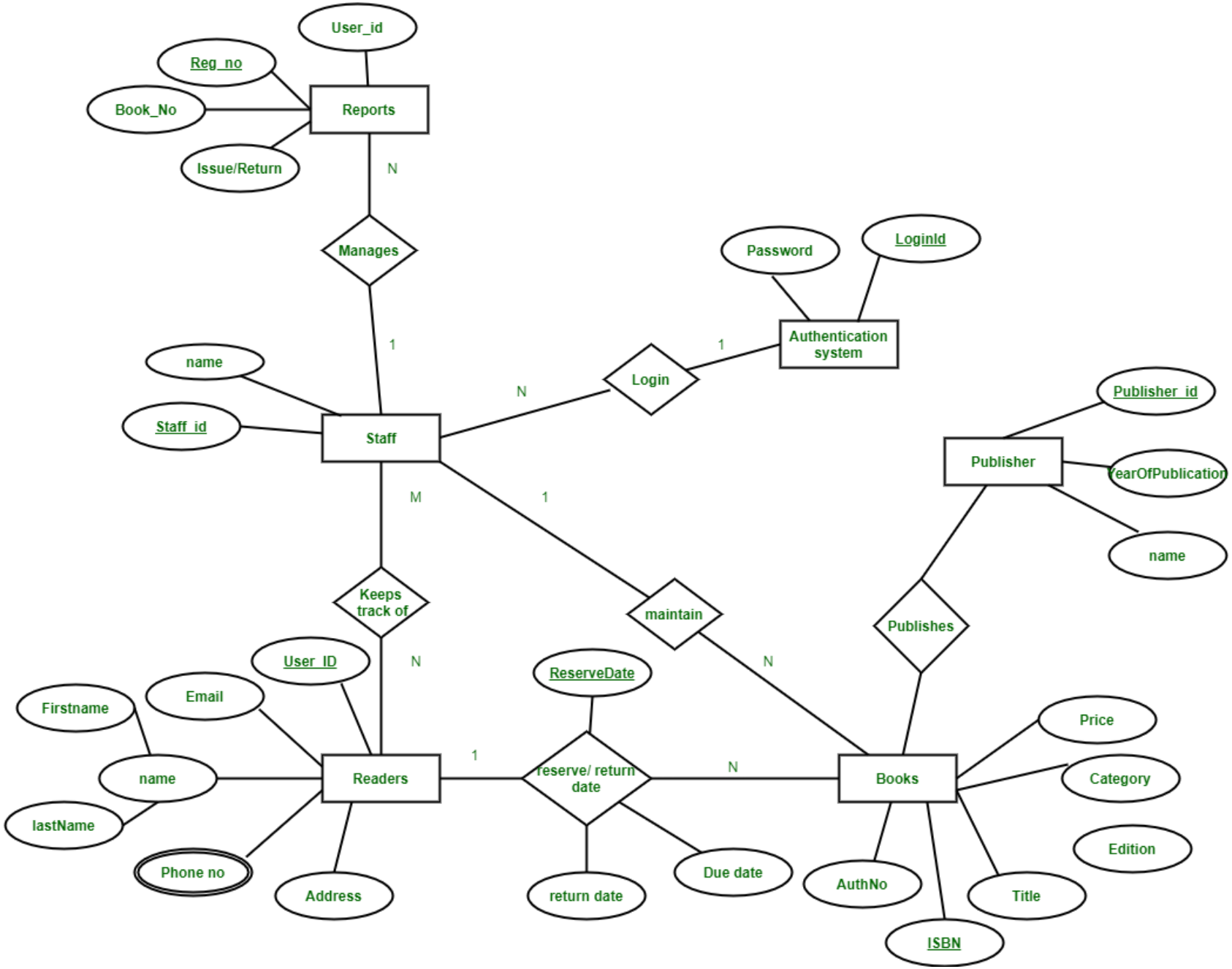
Step 1: Identify the entity types.

Step 2: Identify the relationship types and their participating entity types.

Step 3: Identify the attributes, keys, and identifier of each entity type and relationship type and obtain an ER diagram

Step 4: Convert the ER diagram to a normal form ER diagram.

Examine this ER diagram



Assignment

- Go to course website and check out exercise. Try to built the given exercise diagram by yourself. Use an online tool (e.g. https://app.diagrams.net/) to draw it and print it and bring back to class next week.

THE END

CHECK OUT COURSE WEBSITE: HTTPS://GAZI-END-MUH.GITHUB.IO

GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM